

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Alessandro Gori*

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Abstract

[To be continued...]

- Appendix on s -wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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Contents

1	Mean-field theory discussion of the EHM	1
1.1	Extended Hubbard model symmetries and Wick's theorem	1
1.1.1	Spatial and topological symmetries	1
1.1.2	Spin and charge symmetries	2
1.1.3	Particle-hole symmetry at half-filling	3
1.2	MFT on Hubbard lattices	5
1.2.1	MFT general procedure, self-consistency and Wick's theorem	5
1.2.2	MFT treatment of the pure Hubbard model	6
1.3	The non-local attraction $\hat{H}_V$ as a source of symmetry-breaking interactions	7
1.3.1	Real space description	7
1.3.2	Fourier transform of the model	9
1.4	Iterative computational approach to MFT	11
1.4.1	Sketch of the Hartree-Fock iterative algorithm	11
1.4.2	Self-consistency and RMPs	12
1.4.3	Fixed density simulations and the role of μ	12
1.4.4	The reject-mix-iterate convergence scheme	13
1.5	Square lattice spatial symmetry channels	14
2	The normal phase	19
2.1	The “normal” phase and its symmetries	19
2.2	Hartree effects: chemical potential renormalization	19
2.2.1	Hartree channel in the normal phase: analytic derivation	20
2.2.2	Hartree contraction energy	21
2.3	Fock effects: bare bands renormalization	21
2.3.1	Fock channel in the normal phase: analytic derivation	22
2.3.2	Fock contraction energy	24
2.3.3	Nematicity: extending hopping to d -wave bands	24
2.4	Normal free energy density	26

*a.gori23@studenti.unipi.it / nepero27178@github.com

2.5	Theoretical constraints on the normal phase HFPs	27
2.5.1	Constraints in presence of a robust C_4 symmetry and bands shrinking . . .	29
2.5.2	Natural suppression of nematicity within $C_2/\mathbb{Z}_2$ symmetry	31
2.5.3	Expected dependence on δ at low temperature	33
2.6	Discussion of numeric results	35
2.6.1	Mott localization effect	36
2.6.2	Solution free energy and entropy densities	37
2.6.3	Convergence problems and mixing fine-tuning	39
3	Antiferromagnetic instability	41
3.1	The AF phase and its symmetries	41
3.1.1	Antiferromagnetism in the HM	41
3.1.2	Extension to the EHM	43
3.2	Non-local Hartree effects: chemical potential renormalization	44
3.2.1	Hartree channel in the AF phase: analytic derivation	45
3.2.2	Hartree contraction energy	46
3.3	Fock effects: gap and bands renormalization	46
3.3.1	Fock channel in the AF phase: analytic derivation	47
3.3.2	Fock contraction energy	50
3.4	Full MFT hamiltonian and gap complexification	51
3.4.1	The relative phase in the zero temperature half-filled ground state	52
3.4.2	AF phase flavours and spin degeneracy	52
3.5	Stability of the AF phase	53
3.6	AF free energy density	54
3.6.1	The gap energy	55
3.6.2	Free energy at half-filling	56
3.7	Theoretical constraints on the AF phase HFPs	57
3.7.1	Robust C_4 bands shrinking extension to AF phases	57
3.7.2	HFPs vanishing in sAF phase	58
3.7.3	HFPs vanishing in aAF phase	61
3.7.4	Natural suppression of nematicity within $C_2/\mathbb{Z}_2$ symmetry	62
3.8	Discussion of numeric results	64
3.8.1	Magnetization boost	64
3.8.2	Bands shrinking	65
3.8.3	Solution free energy and entropy densities	67
3.8.4	Fixed-point interpretation of HF solution	67
	Bibliography	69

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

Antiferromagnetic instability

This chapter is devoted to an in-depth discussion of the antiferromagnetic (AF) long-range ordering in the EHM by the means of MFT. In the context of the 2D HM, discussed in App. ??, AF ordering establishes as an unconventional metallic phase which is only stable at half-filling (as is discussed in Sec. ??). The situation for the EHM is similar: the present analysis is carried out based on the argument that, given the symmetry structure of the non-local NN interaction, the fundamental features of the commensurate AF phase of the HM are effectively preserved, while the parameters are redefined by the interaction itself. The two most relevant effects we discuss in this chapter are the gap “complexification” and the bands renormalization and resymmetrization (both in Sec. 3.3), the latter being a general feature of the EHM common to all phases preserving (or, at most, weakening without destroying) translational invariance, discussed previously in Sec. 2.3.

1.1 The AF phase and its symmetries

The AF phase is not uniquely defined; actually, there exists a variety of ways to establish a SDW phase in a spin-balanced square lattice, which means a system where both spin sectors are equally populated, but differently displaced. [\[ToDo: graphics of various SDW structures\]](#)

The AF phase hereby presented is a symmetry reduction of the normal phase, and is described by simultaneous spin rotational symmetry reduction and translational symmetry halving:

$$\begin{aligned} \text{SU}(2) &\xrightarrow{\text{SSB}} \text{U}^z(1) \\ \text{T}_1^{(x)} \otimes \text{T}_1^{(y)} &\xrightarrow{\text{SSB}} \text{T}_2^{(x)} \otimes \text{T}_2^{(y)} \end{aligned}$$

We describe a SDW whose period is of two sites. As done for the normal phase, we allow theoretically

$$\text{C}_4 \xrightarrow{\text{SSB}} \text{C}_2$$

in order to capture the same anisotropy effects seen in Sec. 2.3.1.

Recall Tabs. 1.3 and 1.4, collecting the RWCs for the AF phase,

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate the role of all RWCs in Tab. 3.1.

1.1.1 Antiferromagnetism in the HM

In this short section we recall the principal features of the AF phase established in the conventional HM, discussed in Sec. ?.?. Before starting, let us define as in Eqns. (??) and (??) the AF and density fluctuations operators,

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \tag{1.1}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \tag{1.2}$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
	o.s.	Hartree	μ shift by nzV
$\hat{H}_V$		Hartree	μ shift by nzV
	s.s.	Fock	Bands renormalization and resymmetrization, gap complexification

Table 1.1 | Relevant Wick's contractions and net effect in the AF phase. Note the similarity with Tab. 2.1.

which represent the two essential fermionic quadratic operators of the analytic MFT derivation. It is very useful to also note the following properties. Given the γ -wave form factors of Tab. 1.6, evidently both the following equations hold

$$\begin{aligned}\hat{\psi}_{\mathbf{k}+\boldsymbol{\pi}\sigma}^\dagger &= \hat{\psi}_{\mathbf{k}\sigma} \\ \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)} &= -\varphi_{\mathbf{k}}^{(\gamma)}\end{aligned}$$

which imply the handy property

$$\varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)} \hat{\psi}_{\mathbf{k}+\boldsymbol{\pi}\sigma}^\dagger = -\varphi_{\mathbf{k}}^{(\gamma)} \hat{\psi}_{\mathbf{k}\sigma} \quad (1.3)$$

Within the pure HM, the AF phase is specified by the Ansatz (??),

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}m} \quad (1.4)$$

which is explicitly breaking translational invariance in each spin sector, while preserving $U^z(1)$ and $U^c(1)$ symmetries. As reported in Tab. 1.3, the only RWC comes in Hartree channel, since the local two-body interaction couples operators at opposite spin index. The hamiltonian $\hat{H}$ is reduced thereby to the form of Eq. (??)

$$\begin{aligned}\hat{H}_t + \hat{H}_U \stackrel{\text{MFT}}{\simeq} & -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}] + nU \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] \\ & - mU \sum_{\mathbf{r}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] - E_{\text{H}/U}^{(\text{AF})}\end{aligned}$$

with $E_{\text{H}/U}^{(\text{AF})}$ the contraction energy resulting from double counting terms, given explicitly in Eq. (??),

$$E_{\text{H}/U}^{(\text{AF})} = L_x L_y \times U (n^2 - m^2)$$

In reciprocal space, the hamiltonian decomposes as in Eq. (??),

$$\hat{H}_t + \hat{H}_U \stackrel{\text{MFT}}{\simeq} \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^\dagger h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + nU \times \hat{N} - E_{\text{H}/U}^{(\text{AF})} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma}^* \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

and $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Nambu spinorial formulation is used,

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

and the free electrons energy is simply the tight binding energy of Eq. (1.14) written with an explicit spin index,

$$\epsilon_{\mathbf{k}\sigma} = -2t (\cos k_x + \cos k_y)$$

which is spin-invariant. The MFT description of the model reduces to a gas of free “ γ -fermions”, described by the Nambu spinor of Eq. (??),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix}$$

EV	Meaning	Equation
$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle$	$\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \rangle$	(??)
$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle$	$i \langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^\dagger \rangle$	(??)
$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \rangle$	(??)

Table 1.2 | List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM.

where

$$W_{\mathbf{k}\sigma} = \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} & -\sin \theta_{\mathbf{k}\sigma} \\ \sin \theta_{\mathbf{k}\sigma} & \cos \theta_{\mathbf{k}\sigma} \end{bmatrix} \quad \text{and} \quad \sin 2\theta_{\mathbf{k}\sigma} \equiv \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}$$

These fermions populate the two bands $\pm E_{\mathbf{k}\sigma} = \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2}$. The entire system is mapped onto an ensemble of pseudo-spins, each subject to a pseudo-field, as in Fig. 3.1. To diagonalize the system essentially means to anti-align each pseudo-spin with the z axis. Each pseudospin component expectation value (EV) is described by a simple equation and has a precise meaning, listed in Tab. 3.2.

1.1.2 Extension to the EHM

Consider now the non-local interaction $\hat{H}_V$. Being the RWCs the same as for the normal phase, the discussion is essentially the same of Chap. 2, with the only relevant difference being the Ansatz we use in Eq. (2.1), substituted by the commensurate oscillatory Ansatz of Eq. (3.4). **By itself, even if it may seem the contrary, the Ansatz does not imply C_4 symmetry preservation. It is possible to reduce point group symmetry having still a rotation-symmetric magnetization, as well as it was possible to do in the normal phase with keeping a uniform density.** As for the normal phase, by design the MFT solution to the model for the given phase is described by a redefinition of the various physical quantities and mathematical objects,

$$\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma} \quad E_{\mathbf{k}\sigma} \rightarrow \tilde{E}_{\mathbf{k}\sigma} \quad \Delta_{\mathbf{k}\sigma} \rightarrow \tilde{\Delta}_{\mathbf{k}\sigma} \quad \mu \rightarrow \tilde{\mu}$$

The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}_{\mathbf{k}\sigma}^2 + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

and the following relations hold (the *tilde* sign indicates the renormalized quantity):

$$\langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \quad (1.5)$$

$$\langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \quad (1.6)$$

$$\langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \quad (1.7)$$

equivalent to the ones for the pure HM, Eqns. (??), (??) and (??). Moving to the tilted frame, the z component of the pseudospin is just a thermal superposition of the “up” and “down” states

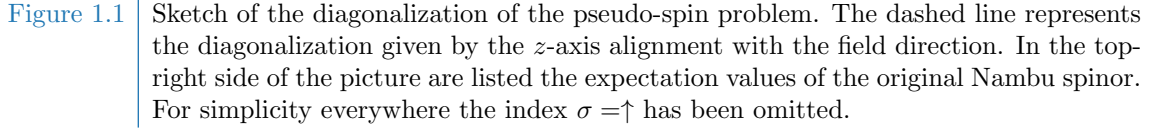
$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) - f(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \quad (1.8)$$

Let us define the (renormalized) thermal factors as in Eq. (??),

$$\tilde{\text{Th}}_{\mathbf{k}\sigma}^{(\pm)}[\beta, \tilde{\mu}] \equiv f(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \pm f(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \quad (1.9)$$

The angles satisfy:

$$\sin 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \quad (1.10)$$



1.2 Non-local Hartree effects: chemical potential renormalization

$$\hat{H}_V \simeq \underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle]}_{\text{s.s.}} - \underbrace{V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle]}_{\text{o.s.}} + (\text{all the rest})$$
$$-V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\sigma} \rangle] - V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\bar{\sigma}} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle] \quad (1.11)$$

as well as the its scalar counterpart

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle \quad (1.12)$$

which accounts for the energy shift to be accounted when performing Wick's contractions.

1.2.1 Hartree channel in the AF phase: analytic derivation

Let us start from the operator of Expr. (3.11). By inserting the Ansatz of Eq. (??), we get

$$\begin{aligned} & \overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{\delta_{\sigma=\uparrow}} \left[(-1)^{x+y} \hat{n}_{\mathbf{r}'\sigma} + (-1)^{x'+y'} \hat{n}_{\mathbf{r}\sigma} \right]}^{\text{s.s.}} \\ & \underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\bar{\sigma}} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[(-1)^{x+y+\delta_{\sigma=\uparrow}} \hat{n}_{\mathbf{r}'\bar{\sigma}} + (-1)^{x'+y'+\delta_{\bar{\sigma}=\uparrow}} \hat{n}_{\mathbf{r}\sigma} \right]}_{\text{o.s.}} \end{aligned}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, evidently

$$\begin{aligned} (-1)^{x'+y'} &= (-1)^{x+y+1} \\ (-1)^{\delta_{\bar{\sigma}=\uparrow}} &= (-1)^{\delta_{\sigma=\uparrow}+1} \end{aligned}$$

We obtain

$$\begin{aligned} & \overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (density)}} - \overbrace{mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (magnetization)}} \\ & \underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}]}_{\text{o.s. (density)}} + \underbrace{mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}]}_{\text{o.s. (magnetization)}} \quad (1.13) \end{aligned}$$

In the above expressions the various contribution have been separated in “density” contributions and “magnetization” contributions. Let us deal with these separately.

Density terms. Consider the s.s. and o.s. (density) terms of Expr. (3.13),

$$-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}] - nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}] \quad (1.14)$$

The discussion is identical to the one of Sec. 2.2.1 for the normal phase, with the renormalization of μ being the same of Eq. (2.5).

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (1.15)$$

Within this equation we are also already implicitly considering the μ shift effect due to $\hat{H}_U$, discussed both in Sec. 2.2 and Sec. ??.

Magnetization terms. The s.s. and o.s. (magnetization) terms of Expr. (3.13) perfectly cancel out:

$$mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [(\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}) - (\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}})] = 0$$

thus the only net effect of the Hartree RWCs affects μ . This is somewhat to be expected: in the context of the HM, the gap function emerges in its purely s -wave structure from this precise computation on Hartree terms. However, the $\hat{H}_V$ non-local interaction cannot provide a pure s -wave potential component. Effects on the gap function are to be observed somewhere else.

Let us now move to the last part of the Hartree effects on the AF phase originated by $\hat{H}_V$, which is, the energy shift due to double counting terms originated by MFT decomposition of the quartic interactions $\hat{H}_U$ and $\hat{H}_V$.

1.2.2 Hartree contraction energy

Consider the scalar Expr. (3.12): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector. In computing the contraction energy from Hartree channel, being the operatorial result identical to the one obtained for the normal phase, we expect the same to happen for double counting terms.

s.s. sector double counting energy. For the s.s. part, explicitating the two spin sectors, we get

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle = \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n + (-1)^{x+y}m)(n + (-1)^{x'+y'}m) \right]}^{\sigma=\uparrow} + \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n - (-1)^{x+y}m)(n - (-1)^{x'+y'}m) \right]}_{\sigma=\downarrow}$$

Evidently the mixed terms (those of order $\mathcal{O}(n) \times \mathcal{O}(m)$) cancel out, and considering that the sign factor is necessarily alternating on neighboring site, the remainder is just

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 - m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle = \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n + (-1)^{x+y}m)(n - (-1)^{x'+y'}m) \right]}^{\sigma=\uparrow} + \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n - (-1)^{x+y}m)(n + (-1)^{x'+y'}m) \right]}_{\sigma=\downarrow}$$

which gives immediately

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 + m^2)$$

Hence, the total “contraction energy”, given by the sum of both contributions, is identical to the normal one of Eq. (2.7) indeed:

$$E_{\text{H}/V}^{(\text{AF})} = -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 = 4n^2V \times \frac{z}{2}L_xL_y \quad (1.16)$$

the minus sign coming from the fact that we are subtracting this energy shift.

1.3 Fock effects: gap and bands renormalization

We explored thoroughly the Hartree contractions effects on the MFT hamiltonian. Let us now move to the effects originated by Fock contractions. We can use the very beginning of Sec. 2.3 derivation, up to Eq. (2.9),

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} \rangle \right] \\ \stackrel{\mathcal{F}}{=} \frac{2V}{L_xL_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E_{\text{F}/V}^{(\text{N})} \quad (1.17)$$

where $E_{F/V}^{(\text{AF})}$ is defined as in Eq. (2.10), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

1.3.1 Fock channel in the AF phase: analytic derivation

The commensurate AF phase is essentially described by two EVs being in principle non-zero,

$$|\langle \hat{n}_{\mathbf{k}\sigma} \rangle| \geq 0 \quad |\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle| \geq 0$$

Because of this, for the conventional HM the AF ground state is a degenerate Fermi gas of $\gamma^{(\pm)}$ quasiparticles, and for null temperature acquires the simple form of Eq. (??). Assuming the new ground state to be similar and going back to Eq. (3.17), this implies only two contributions are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{or} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}'$$

The first case gives us the normal fermionic EV, $\langle \hat{n}_{\mathbf{k}\sigma} \rangle$, the second the staggered AF order parameter, $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle$. Then Eq. (3.17) is reduced to:

$$(\dots) \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \left[\underbrace{\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \right] \quad (1.18)$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (3.18) are simple density interactions with the mean density field. Consider Fig. 2.1a: density at vector $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$ interacts with the mean density at vector $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. These variables are depicted in Fig. 2.1b. Apply in the diagonal part of Eq. (3.18) this variables change (which is the same of Eq. (2.13))

$$\begin{aligned} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ = \frac{2V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \\ = \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{n}_{\mathbf{q}'\sigma} \end{aligned} \quad (1.19)$$

where in the last passage we substituted the decomposition of Eq. (1.27). Define as in Eq. (2.15) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \quad (1.20)$$

The expression in square brackets of Eq. (3.19) is just its γ component, and is expanded in

$$\begin{aligned} u_{\sigma}^{(\gamma)} &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} [\langle \hat{n}_{\mathbf{q}\sigma} \rangle - \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}_{\mathbf{q}\sigma}^\dagger \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\text{Th}}_{\mathbf{q}\sigma}^{(-)}[\beta, \tilde{\mu}] \end{aligned} \quad (1.21)$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4b
$d_{x^2-y^2}$ -wave	$u^{(d)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4e

Table 1.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

where, in the last passage, Eqns. (3.7), (3.8) and (3.9) have been used. Now, the AF phase we realize does not break inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalized bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \implies u_{\sigma}^{(p_\ell)} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

The renormalized bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. The remaining s^* and d symmetries are legitimate HFPs with relative self-consistency equations, reported in Tab. 3.3 using spin degeneracy.

Recognizing the harmonics, Eq. (3.19) becomes

$$\begin{aligned} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ = V \sum_{\mathbf{q}, \sigma} \left[u_{\sigma}^{(s^*)} (\cos q_x + \cos q_y) + u_{\sigma}^{(d)} (\cos q_x - \cos q_y) \right] \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} \end{aligned}$$

These two factors, together, define the full bands renormalization

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_{\sigma}^{(s^*)} V (\cos k_x + \cos k_y) + u_{\sigma}^{(d)} V (\cos k_x - \cos k_y) \quad (1.22)$$

as was for the normal phase in Eq. (2.17). In Sec. 2.3.3 we discussed the appearance of anisotropic hopping with d -wave extension of the bands: here the theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Off-diagonal terms. Consider the off-diagonal terms of Eq. (3.18). These contribute instead to the full gap, being out of diagonal in the 2×2 hamiltonian matrix. In terms of the pseudospin picture, they account for a transverse field. Define $\mathbf{q}, \mathbf{q}'$ as in Fig. 2.1b,

$$\mathbf{q} \equiv \mathbf{K} + \mathbf{k} \quad \mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$$

and rewrite as for Eq. (3.19)

$$\begin{aligned} -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\ = -\sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma}^\dagger \rangle \right] \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}_{\mathbf{q}'\sigma}^\dagger \end{aligned}$$

Now shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k} + \boldsymbol{\pi} \equiv \mathbf{q}$, and use Eq. (3.3) as well as reciprocal space periodicity by a BZ,

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma}^\dagger \rangle &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}+\boldsymbol{\pi}\sigma}^\dagger \rangle \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \end{aligned}$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$v^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4b
p_x -wave	$v^{(p_x)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \sin k_x \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4c
p_y -wave	$v^{(p_y)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \sin k_y \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4d
$d_{x^2-y^2}$ -wave	$v^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x - \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. 1.4e

Table 1.4 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_{\mathbf{k}}$.

which finally gives

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}_{\mathbf{q}'\sigma}^{\dagger} \quad (1.23)
\end{aligned}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \quad (1.24)$$

(which is just the AF order parameter with a complex factor) and its γ -wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned}
v_{\sigma}^{(\gamma)} &= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\pi}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}+\pi\sigma} \rangle \right] \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \left[\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle - \langle \hat{\psi}_{\mathbf{q}\sigma}^{\dagger} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}_{\mathbf{q}\sigma}^{\dagger} \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\text{Th}}_{\mathbf{q}\sigma}^{(-)}[\beta, \tilde{\mu}] \quad (1.25)
\end{aligned}$$

where we made use of Eq. (3.3), the relations of Tab. 3.2 as well as Eq. (3.6). There is no particular reason to exclude one component or another. In Tab. 3.4 the four derived self-consistency equations are reported (already ignoring spin index).

From the definitions of the structure factors of Tab. 1.6, it's immediate to see that:

$$v_{\sigma}^{(s^*)}, v_{\sigma}^{(d)} \in \mathbb{R} \quad v_{\sigma}^{(p_x)}, v_{\sigma}^{(p_y)} \in i\mathbb{R}$$

Getting back to Eq. (3.23) and inserting these components, we get

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = -iV \sum_{\gamma, \sigma} v_{\sigma}^{(\gamma)} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^{\dagger} \quad (1.26)
\end{aligned}$$

We can expand on the MBZ as

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = -iV \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^\dagger + v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}+\pi}^{(\gamma)*} \hat{\psi}_{\mathbf{q}+\pi\sigma}^\dagger \right]
\end{aligned}$$

and use Eq. (3.3), exchanging the terms positions

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = iV \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma} - v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^\dagger \right] \quad (1.27)
\end{aligned}$$

Now: recalling Tab. 1.6, for the four symmetries we are considering, the form factors are real for symmetries s^* and d , and purely imaginary for p_x and p_y . As we mentioned before, the same holds for the $v_{\sigma}^{(\gamma)}$ harmonics. Then

$$i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \in i\mathbb{R} \quad (1.28)$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} = \left(i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \right)^*$$

Therefore, we conclude

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = V \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma} + \text{h.c.} \right] \quad (1.29)
\end{aligned}$$

which is obvious, being the hamiltonian operator hermitian from the beginning.

1.3.2 Fock contraction energy

Let us now get back to $E_{\text{F}/V}^{(\text{AF})}$, the double counting energy of Eq. (3.17). As said before, its definition is the same as for the normal phase, Eq. (2.10), with the averaging operation being performed on the AF ground state:

$$E_{\text{F}/V}^{(\text{AF})} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \quad (1.30)$$

Proceeding exactly as in Eq. (3.18) and making the change of variables of Fig. 2.1, we get

$$E_{\text{F}/V}^{(\text{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \left[\underbrace{\langle \hat{n}_{\mathbf{q}\sigma} \rangle \langle \hat{n}_{\mathbf{q}'\sigma} \rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \langle \hat{\psi}_{\mathbf{q}'\sigma} \rangle}_{\text{Off-diagonal terms}} \right]$$

Apply the decomposition of Eq. (1.27) and use Eqns. (3.20), (3.24),

$$E_{\text{F}/V}^{(\text{AF})} = \frac{V}{2L_x L_y} \sum_{\gamma, \sigma} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} [u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} + v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma}] \quad (1.31)$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma} \rangle \implies u_{\mathbf{q}\sigma} = u_{\mathbf{q}\sigma}^*$$

and then

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\ &= |u_{\sigma}^{(\gamma)}|^2 \end{aligned}$$

At the same time, being $\hat{\psi}_{\mathbf{q}+\pi\sigma} = \hat{\psi}_{\mathbf{q}\sigma}^\dagger$,

$$v_{\mathbf{q}\sigma} = i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \implies v_{\mathbf{q}+\pi\sigma} = -v_{\mathbf{q}\sigma}^*$$

Since the form factors satisfy $\varphi_{\mathbf{q}+\pi}^{(\gamma)} = -\varphi_{\mathbf{q}}^{(\gamma)}$, we get:

$$\begin{aligned} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} v_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}'\sigma} \right] \\ &= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}+\pi}^{(\gamma)*} v_{\mathbf{k}+\pi\sigma} \right] \\ &= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)*} v_{\mathbf{k}\sigma}^* \right] \\ &= |v_{\sigma}^{(\gamma)}|^2 \end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k} + \pi$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \text{BZ}$. Inserting the above results in Eq. (3.31), this last becomes:

$$E_{\text{F}/V}^{(\text{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma, \sigma} \left[|u_{\sigma}^{(\gamma)}|^2 + |v_{\sigma}^{(\gamma)}|^2 \right] \quad (1.32)$$

Interestingly enough, the new HFPs amplitudes contribute positively to the contraction energy. This fact is non-trivial and has consequences we will discuss in Sec. 3.6.1.

1.4 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The MFT model is approximated by:

$$\begin{aligned} \hat{H} - \mu \hat{N} &\simeq \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} [\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\pi\sigma}] - \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right] \\ &\quad - \tilde{\mu} \hat{N} - \left[E_{\text{H}/U}^{(\text{AF})} + E_{\text{H}/V}^{(\text{AF})} + E_{\text{F}/V}^{(\text{AF})} \right] \end{aligned}$$

where the renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma \in \{s^*, d\}} u_{\sigma}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} - iV \sum_{\gamma} v_{\sigma}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \quad (1.33)$$

The gap correction is purely imaginary, compared to the HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. This has two obvious consequences: being no other contribution to the x pseudospin projections other than the pure HM ones, spin antisymmetry is preserved within such direction

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = -\langle \hat{s}_{\mathbf{k}\bar{\sigma}}^x \rangle$$

which is relevant in the computation of the physical magnetization, which (as is proven in Eq. (??)) is defined

$$\begin{aligned} m &= \frac{1}{2L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{s}_{\mathbf{k}\uparrow}^x - \hat{s}_{\mathbf{k}\downarrow}^x \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{s}_{\mathbf{k}\uparrow}^x \rangle \end{aligned}$$

A proof of the above derivation is given in Sec. ???. Moreover, the net effect $\hat{H}_V$ is a “gap complexification”. In other words,

$$|\text{Im}\{\Delta_{\mathbf{k}\sigma}\}| \geq 0 \quad \implies \quad |\langle s_{\mathbf{k}\sigma}^y \rangle| \geq 0$$

while for the HM the y pseudospin component was identically null.

1.4.1 The relative phase in the zero temperature half-filled ground state

In Sec. ??, we derive analytically the AF ground state on the conventional HM at zero temperature, as a simple Bloch’s transform of a polarized (tilted) pseudospins collective pure state. Eq. (??) reports the final form of said ground-state. Said quantum state is in general pretty similar to the conventional BCS state.

At half-filling and zero temperature, it is rather easy to extend that state to the new complexified gap. By performing an identical calculation as in Sec. ?? we here omit, it’s easy to show that the new ground state for the EHM in the AF phase is identical, with RMPs in the various amplitudes and a relative phase taking care of the gap function complex phase,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (1.34)$$

which is identical to Eq. (??), apart from renormalizations. Let $|\mathbf{k}\sigma\rangle \equiv \hat{c}_{\mathbf{k}\sigma}^{\dagger} |\Omega_{\mathbf{k}\sigma}\rangle$. Then we can write

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} \right] |\mathbf{k}\sigma\rangle$$

having we used $|\Omega_{\mathbf{k}\sigma}\rangle = \{\hat{c}_{\mathbf{k}\sigma}, \hat{c}_{\mathbf{k}\sigma}^{\dagger}\} |\Omega_{\mathbf{k}\sigma}\rangle$.

The angles appearing in the above equations are the same of Eq. (3.10), which acquire a physical meaning: the relative phase $2\tilde{\zeta}_{\mathbf{k}\sigma}$ is the gap function angle in complex plane, since

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}\sigma}^x | \Psi \rangle &= \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \Psi \rangle \\ &= \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}\sigma}^y | \Psi \rangle &= i \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \Psi \rangle \\ &= i \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} - e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

These equations are coherent with Eqns. (3.5), (3.6), thus the involved angles apart from irrelevant 2π shifts coincide.

1.4.2 AF phase flavours and spin degeneracy

The AF phase we are studying is not allowed to break inversion symmetry, a core property of the system which is respected from all our approximations. This translates in the simple requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

Phase	HFPs
sAF	$\{m, u^{(s*)}, u^{(d)}, v^{(s*)}, v^{(d)}\}$
aAF	$\{m, u^{(s*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 1.5 | Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}_{\mathbf{k}\sigma}^2 + \text{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \text{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

The renormalized bare bands as well as the real part of the gap satisfy the inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \quad \text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \text{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm \text{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this trivial consideration, we are able to discard two of the four self-consistency equations for the v -related HFPs listed in Tab. 3.4: s^* and d -wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y -wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors. We name the two possible symmetries “sAF” (symmetric antiferromagnet) and “aAF” (antisymmetric antiferromagnet). The set of HFPs then restricts in these AF structures, and is reported for both cases in Tab. 3.5. The presence itself of two “AF flavours” is not-so-physical, given that up to now we have no clear physical interpretation for $\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$. As we will discuss in Sec. 3.7, it is possible to prove that almost all these new HFPs actually vanish – a fact that is not inferable from MFT analysis, but can be seen only delving into the detailed discussion of the iterative algorithm seen as a recursive parametric map. For now, let us proceed integrally with all the HFPs we found and derive the results.

Moreover, from now on we will omit the σ index, working just in the $\uparrow$ sector. As said in Sec. 3.1, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but globally symmetric spin density distribution among sites. The MFT hamiltonian is invariant by a $\sigma \leftrightarrow \bar{\sigma}$ exchange, thus all measurable physical quantities must be. What is unmeasurable and has no effect on Physics is free to be σ -dependent, as for the sign of $\text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$. Since all it’s relevant of both the complex components of the gap function is their magnitude, we can safely treat σ as a pure degeneracy. This gives a simple equation for the last HFP,

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{s}_{\mathbf{k}\uparrow}^x \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \end{aligned} \quad (1.35)$$

which is the actual physical magnetization. From the first line we see that it can be interpreted as the average transverse pseudospin value.

1.5 Stability of the AF phase

[Re-write and move up.]

The discussion above is formally correct and self-consistent, thus a HF algorithm can be sketched to extract the HFPs. However we cannot proceed without considering the instability effects of Sec. ???. As is known (and we would have liked to notice earlier) the AF-SSB at wavevector π we are studying here, responsible of halving the BZ down to the MBZ, can produce a stable phase on pure Hubbard lattices only at half filling [15]. For the EHM the situation is even

worse than that: repeating the simple Linear Response Theory (LRT) computations of Sec. ?? and mimicking Eq. (??), we get a proper response function

$$\tilde{\chi}_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\mathbf{k}, \omega) = \frac{\chi_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\mathbf{k}, \omega)}{1 - [U - 2V(\cos k_x + \cos k_y)] \chi_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\mathbf{k}, \omega)} \quad (1.36)$$

[Double check the red part and conclude this paragraph...].

As is well documented in literature [17], AF can establish in a wide range of spatial structures with particular shapes, for example stripes. In order to extract the precise AF phase, we should

1. Choose an initializer wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve self-consistently for the HFPs at $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\nabla F < 0$;
5. If the gradient descent is found, update $\mathbf{k} \rightarrow \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

Of course the sketched procedure can lead to a lot of complications, first the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions.

In this text we will show the self consistent result for AF-SSB at wavevector $\boldsymbol{\pi}$ even at finite doping as a first step in the above sketched procedure. The presented results are self-consistent, but unfortunately deeply unstable and unphysical at finite doping, a situation where an incommensurate and insulating AF phase is preferred to this commensurate metallic counterpart. Nevertheless we present them anyway.

Let us gather some more informations on the HFPs we identified. [To be continued...]

1.6 AF free energy density

In order to derive the expression of the free energy density for the AF phase, we proceed precisely as in Sec. 2.4. The only notable difference is in entropy: Eq. (2.21) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})\right) \ln \left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})\right) + f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \quad (1.37)$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu \hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left[(\tilde{E}_{\mathbf{k}} - \tilde{\mu}) \hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right] + E_c^{(\text{AF})}$$

being $E_c^{(\text{AF})}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. 2.4, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e_g^{(\text{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. Contraction energy:

$$e_c^{(\text{AF})} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \left(|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right) \right]$$

3. Entropic energy

$$\begin{aligned} e_s^{(\text{AF})} = & \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ & + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

4. Chemical potential correction

$$e_n^{(\text{AF})} = \mu n$$

Compose everything

$$f_{\text{MFT}}^{(\text{AF})} \equiv e_g^{(\text{AF})} + e_c^{(\text{AF})} + e_s^{(\text{AF})} + e_n^{(\text{AF})}$$

and we get

$$\begin{aligned} f_{\text{MFT}}^{(\text{AF})} = & \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) \right] \\ & + Um^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right] + n [\mu + n(2zV - U)] \end{aligned}$$

As was for the normal phase, the contraction energy and the chemical potential energy corrections build up the RMP $\tilde{\mu}$ of Eq. (3.15),

$$\begin{aligned} f_{\text{MFT}}^{(\text{AF})} = & \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) \right] \\ & + Um^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right] + \tilde{\mu} n \quad (1.38) \end{aligned}$$

1.6.1 The gap energy

It is useful to see here a property of the above expression. Collecting the terms directly related to the gap function, the following identity holds:

$$Um^2 - V \sum_{\gamma} |v^{(\gamma)}|^2 = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \quad (1.39)$$

The proof is simple and immediate: developing the right-hand side,

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] = & \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ & + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \end{aligned}$$

Now:

$$\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\} = Um$$

which implies

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] = & Um \times \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ = & Um^2 \end{aligned}$$

having we recognized the magnetization from Eq. (3.35). Equivalently,

$$\begin{aligned}\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\} &= -V \sum_{\gamma} v^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \\ &= -V \sum_{\gamma} v^{(\gamma)*} \varphi_{\mathbf{k}}^{(\gamma)}\end{aligned}$$

where we used Eq. (3.28) to exchange the conjugation operation in the last passage. This implies:

$$\begin{aligned}\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] &= -V \sum_{\gamma} v^{(\gamma)*} \sum_{\mathbf{k} \in \text{MBZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ &= -V \sum_{\gamma} v^{(\gamma)*} \times v^{(\gamma)} \\ &= -V \sum_{\gamma} |v^{(\gamma)*}|^2\end{aligned}$$

where in second passage we recognized $v^{(\gamma)}$ given in Eq. (3.25). Then Eq. (3.39) is verified. The latter is particularly interesting because it shows a non-intuitive result: due to gap complexification, gap amplitude is only increased by $\hat{H}_V$, thus by analogy we would expect the relative free energy contribution to have the same sign as the original gap energy from the real part. This is not the case: to open a gap in the s -wave real sector contributes positively to free energy, while to open a gap in the γ -wave imaginary sector contributes negatively.

An immediate consequence of Eq. (3.39) is that, being the right-hand side evidently positive, for $V > 0$ the total amplitude of the “complexifiers” vector is maximised,

$$\sum_{\gamma} |v^{(\gamma)}|^2 \leq M \quad \text{being} \quad M^{(\text{AF})} \equiv m^2 \frac{U}{V} \quad (1.40)$$

The above constraint predicts that:

- In the strong coupling limit $U \ll V$, gap complexification is suppressed;
- Correctly, when the physical magnetization closes, so does the imaginary part of the gap

$$\lim_{m \rightarrow 0} v^{(\gamma)} = 0$$

which is coherent with the entire derivation and takes us back to the normal phase.

Eq. (3.39) is also solved by a null magnetization and thus null complexifiers. This can be interpreted by the fact that when the constraint is impossible to solve self-consistently, the gap must close.

[Graphic interpretation: both the aAF and sAF phases have two-syms *vs*, thus the constraint above tells us that the relative two-elements vector is contained in a circumference.]

1.6.2 Free energy at half-filling

It's of particular interest the half-filling condition, with $\tilde{\mu} = 0$, where PHS discussed in Sec. 1.1.3 is preserved in the AF phase. Since, as it can be shown easily,

$$\begin{aligned}\ln\left(1 - \frac{1}{e^x + 1}\right) + \ln\left(1 - \frac{1}{e^{-x} + 1}\right) &= \ln\left[\frac{e^x}{(e^x + 1)^2}\right] \\ &= x - 2\ln(1 + e^x)\end{aligned}$$

Let now $x = \beta \tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $x \rightarrow +\infty$, which gives

$$x - 2\ln(1 + e^x) \simeq -x$$

Hence we get the half-filling free energy density substituting ,

$$f_{\text{MFT}}^{(\text{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \tilde{E}_{\mathbf{k}} + U m^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right]$$

Starting from the full expression of the free energy density, we recovered the intuitive expression of the energy of a fully (double) occupied band at $-\tilde{E}_{\mathbf{k}}$, with HFPS correction.

[Expand: interesting situation for bands with nodes. At low temperature, where band nodes are present, the function behaves differently because its argument goes to zero. What is the behaviour there?]

1.7 Theoretical constraints on the AF phase HFPS

Up to now, all we derived was the most general MFT-compatible mathematical description of a commensurate AF established on the EHM, allowing for nematic spatial symmetry breaking as well as non-trivial renormalization effects. This section, a clone of Sec. 2.5, is dedicated to understanding, given the algorithmic problem, how its solutions should behave. As we will see, based on this theoretical analysis we are able to reduce a lot the algorithmic complexity and limit the solution search to a tailored subspace of HFPS space.

We start from considering a robust C_4 antiferromagnet, thus setting $u^{(d)} = 0$ by hand, and show how the constraints on $u^{(s^*)}$ mimic those posed for the normal phase in Eq. (2.24). Then we move to a rather convoluted but necessary analysis of the five components HFPS vector described for both AF flavours in Tab. 3.5. For the sake of generality, even if – as discussed in Sec. 3.5 – at finite doping the system is naturally unstable, we take on all calculations at generic $\tilde{\mu}$.

1.7.1 Robust C_4 bands shrinking extension to AF phases

For the normal phase, we saw that when C_4 symmetry is preserved (Sec. 2.5.1) we can conclude immediately the bands renormalization is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)} V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

As we did for the normal phase:

1. If $t > 0$ and $u^{(s^*)} \leq 0$

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0$$

the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives

$$u^{(s^*)} \leq 0 \quad \implies \quad u^{(s^*)} \geq 0$$

But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily

$$u^{(s^*)} > 0$$

2. If $t > 0$ and $u^{(s^*)} \geq 2t/V$

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$

the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives

$$u^{(s^*)} \geq 2t/V \quad \implies \quad u^{(s^*)} \leq 0$$

We conclude necessarily

$$u^{(s^*)} < 2t/V$$

The final result,

$$0 < u^{(s^*)} < 2t/V \quad (1.41)$$

is identical to the constraint obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\hat{H}_V$.

1.7.2 HFPs vanishing in sAF phase

We now present a somewhat convoluted argument to show that, even if raw MFT presents an apparently rich scenario with a lot of HFPs, the problem can be proven to be actually much simpler. Let us start from the sAF phase, assuming

$$C_4 \xrightarrow{\text{SSB}} C_2$$

The system is described by the five HFPs on the first row of Tab. 3.5 and their self-consistency equations of Tabs. 3.3 and 3.4.

Let us define the functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k}) \quad (1.42)$$

whose linearity properties are trivial,

$$F[af(\mathbf{k}) + bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})] \quad (1.43)$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} : f(\mathbf{k}) \geq 0 \implies F[f(\mathbf{k})] \geq 0 \quad (1.44)$$

The relation above is implied by the fact that both $\tilde{\text{Th}}_{\mathbf{k}}^{(-)}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined,

$$\tilde{\text{Th}}_{\mathbf{k}}^{(-)} = \frac{1}{e^{\beta(\tilde{E}_{\mathbf{k}} - \mu)} + 1} - \frac{1}{e^{\beta(-\tilde{E}_{\mathbf{k}} - \mu)} + 1} \geq 0 \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \geq 0$$

so, being the functional F essentially a weighted integral over MBZ, any positive-defined function has a positive image. An identical statement with inverted inequalities holds for negative-defined function.

Consider now the self-consistency equation for $u^{(s^*)}$: explicitly,

$$\begin{aligned} u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (c_x + c_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\ &= \frac{[2t - Vu^{(s^*)}]}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{Vu^{(d)}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2) \end{aligned}$$

having we used the notation $c_\ell \equiv \cos k_\ell$ and substituted the full bare bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u_\sigma^{(s^*)}V\right] (c_x + c_y) + u_\sigma^{(d)}V(c_x - c_y)$$

Evidently, in terms of the functional S , we can rewrite:

$$\begin{aligned} u^{(s^*)} &= \left[2t - Vu^{(s^*)}\right] \times F[(c_x + c_y)^2] - Vu^{(d)} \times F[c_x^2 - c_y^2] \\ &= \left[2t - Vu^{(s^*)}\right] \times \{F[c_x^2] + 2F[c_x c_y] + F[c_y^2]\} - Vu^{(d)} \times \{F[c_x^2] - F[c_y^2]\} \end{aligned}$$

having we used the linearity property of Eq. (3.43) in last passage.

Let now:

$$A \equiv F[c_x^2] \quad B \equiv 2F[c_x c_y] \quad C \equiv F[c_y^2] \quad (1.45)$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)} \right] (A + B + C) - Vu^{(d)}(A - C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)} \right] (A + B + C) - Vu^{(d)}(A - C) \quad (1.46)$$

$$u^{(d)} = \left[2t - Vu^{(s^*)} \right] (A - C) - Vu^{(d)}(A - B + C) \quad (1.47)$$

$$v^{(s^*)} = -Vv^{(s^*)}(A + B + C) - Vv^{(d)}(A - C) \quad (1.48)$$

$$v^{(d)} = -Vv^{(s^*)}(A - C) - Vv^{(d)}(A - B + C) \quad (1.49)$$

These equations define a four-fold nonlinear map, whose parameters A, B, C depend on the HFPS themselves, being the functional F dependent on the HFPS. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps.

Aim of this section is to prove that no self-consistent solution exists with non-zero $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$. Being the derivation convoluted and abstract enough, let us proceed by enumerating the relevant facts.

1. Let us start by applying a conformal transform on the map: define the new “sum” and “difference” variables

$$S_u \equiv u^{(s^*)} + u^{(d)} \quad D_u \equiv u^{(s^*)} - u^{(d)} \quad S_v \equiv v^{(s^*)} + v^{(d)} \quad D_v \equiv v^{(s^*)} - v^{(d)}$$

whose matrix representation is

$$\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ & 1 & 1 \\ & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (3.46) and (3.47) (the first two) as well as Eq. (3.48) and (3.49) (the second two). By doing this we get the simplified system:

$$(1 + 2AV)S_u + BVD_u = 2t(2A + B) \quad (1.50)$$

$$(1 + 2CV)D_u + BVS_u = 2t(B + 2C) \quad (1.51)$$

$$(1 + 2AV)S_v + BVD_v = 0 \quad (1.52)$$

$$(1 + 2CV)D_v + BVS_v = 0 \quad (1.53)$$

2. Let $S_v = 0$. Then $D_v = 0$. From Eq. (3.53), using $C = F[c_y^2] \geq 0$ implied by Eq. (3.44):

$$\underbrace{(1 + 2CV)}_{>0} D_v = 0 \quad \implies \quad D_v = 0$$

Identically, let $D_v = 0$. Then applying the same consideration on Eq. (3.52), $S_v = 0$. Either both are non-zero or both are zero.

3. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B \neq 0$. To prove this, consider for example Eq. (3.52): we are working at $V > 0$, thus

$$B = -\frac{1 + 2AV}{V} \frac{S_v}{D_v}$$

which is non-zero because $A = F[c_x^2] \geq 0$, the latter following from Eq. (3.44). We could have used identically Eq. (3.53).

4. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$(BV)^2 = (1 + 2CV)(1 + 2AV) \quad (1.54)$$

To prove the latter, it is sufficient to multiply Eq. (3.52) by $BV \neq 0$ (see point above),

$$BV(1 + 2AV)S_v + (BV)^2 D_v = 0$$

and substitute from Eq. (3.53) $BVS_v = -(1 + 2CV)D_v$,

$$[-(1 + 2CV)(1 + 2AV) + (BV)^2] D_v = 0$$

Since $D_v \neq 0$ by hypothesis, Eq. (3.54) is implied.

5. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$B + 2C = V(B^2 - 4AC) \quad (1.55)$$

To prove this: multiply Eq. (3.50) by BV on both sides,

$$BV(1 + 2AV)S_u + (BV)^2 D_u = 2tBV(2A + B)$$

and substitute from Eq. (3.51) $BVS_v = -(1 + 2CV)D_v + 2t(B + 2C)$,

$$-(1 + 2CV)(1 + 2AV)D_v + 2t(B + 2C)(1 + 2AV) + (BV)^2 D_u = 2tBV(2A + B)$$

$$2t(B + 2C)(1 + 2AV) = 2tBV(2A + B)$$

$$B + 2ABV + 2C + 4ACV = 2ABV + B^2V$$

Passing from first to second line we used Eq. (3.54), and in the passage below $t > 0$. Last line implies Eq. (3.55).

6. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B^2 \neq 4AC$. Indeed, let $B^2 = 4AC$. Then from Eq. (3.54),

$$(BV)^2 = 1 + 2V(A + C) + 4ACV^2 \implies 1 + 2V(A + C) = 0$$

which is obviously absurd, being $A \geq 0$, $C \geq 0$ from Eq. (3.44). A consequence of this implication on Eq. (3.55) is that, at a given V , the map parameters should satisfy

$$V = \frac{B + 2C}{B^2 - 4AC} \quad (1.56)$$

7. Let $S_v \neq 0$ and $D_v \neq 0$. Then $A = C$. To prove this, substitute Eq. (3.56) in Eq. (3.54),

$$(B^2 - 4AC) \left(\frac{B + 2C}{B^2 - 4AC} \right)^2 - 2 \frac{B + 2C}{B^2 - 4AC} (A + C) - 1 = 0$$

$$(B + 2C)^2 - 2(B + 2C)(A + C) - (B^2 - 4AC) = 0$$

$$B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2 - B^2 + 4AC = 0$$

Simplifying all recurring terms, we get

$$2B(C - A) = 0$$

From previous points we know $B \neq 0$. Then $A = C$.

8. It's impossible to have $S_v \neq 0$ and $D_v \neq 0$. To prove this, assume the contrary and use the point above to reduce Eq. (3.56) to

$$A = C \implies V = \frac{B + 2C}{B^2 - 4C^2} = \frac{1}{B - 2C}$$

Inserting the latter in Eq. (3.53),

$$\begin{aligned} \left(1 + \frac{2C}{B-2C}\right) D_v + \frac{B}{B-2C} S_v &= 0 \\ \frac{B}{B-2C} D_v + \frac{B}{B-2C} S_v &= 0 \end{aligned}$$

which implies

$$D_v + S_v = 0 \implies v^{(s^*)} = 0$$

Recalling from Tab. 3.4 the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -V v^{(d)} F[(c_x - c_y)^2]$$

and using the positivity rule of Eq. (3.44) we get

$$V > 0 \quad \text{and} \quad F[(c_x - c_y)^2] \geq 0 \implies v^{(d)} = 0$$

because no finite positive number can equal any finite negative number, and viceversa. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \implies v^{(s^*)} = v^{(d)} = 0 \implies S_v = D_v = 0$$

Absurd. We explored any possibility for S_v, D_v : no stable solution with finite values of the HFPS $v^{(s^*)}, v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

To gain full knowledge on the map fixed point's shape, we should analyze similarly the behaviour of the self-consistency equations for $u^{(s^*)}, u^{(d)}$. This is done later in Sec. 3.7.4. For now, it suffices to know that limitedly to the sAF phase no gap complexification aforementioned in Sec. 3.4 actually occurs. This effect is not a consequence of MFT derivation, which remains self-coherent and correct; it is instead of the particular form of the non-linear map defined by the set of model equations.

1.7.3 HFPS vanishing in aAF phase

We now deal with a much easier proof for an analogous thesis on the sAF phase described by the five HFPS on the second row of Tab. 3.5, as well as the self-consistency equations of Tabs. 3.3 and 3.4. We continue using the functional F as defined in Eq. (3.42). We also define A, B, C as in Eq. (3.45). Let $s_\ell \equiv \sin k_\ell$, and

$$P \equiv F[s_x^2] \quad Q \equiv 2F[s_x s_y] \quad R \equiv F[s_y^2] \quad (1.57)$$

Using these definitions, four of five self-consistency equations are given by

$$u^{(s^*)} = [2t - V u^{(s^*)}] (A + B + C) - V u^{(d)} (A - C) \quad (1.58)$$

$$u^{(d)} = [2t - V u^{(s^*)}] (A - C) - V u^{(d)} (A - B + C) \quad (1.59)$$

$$v^{(p_x)} = -V v^{(p_x)} P - V v^{(p_y)} Q \quad (1.60)$$

$$v^{(p_y)} = -V v^{(p_x)} Q - V v^{(p_y)} R \quad (1.61)$$

To prove that both $v^{(p_\ell)} = 0$ is immediate.

1. For the aAF phase, $Q = 0$. Let us write Q explicitly

$$\begin{aligned} Q &= F[s_x s_y] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\ &= \frac{1}{2L_x L_y} \sum_{k_x = -\pi}^{+\pi} \sin k_x \sum_{k_y = -\pi}^{+\pi} \frac{\tilde{\text{Th}}_{(k_x, k_y)}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{(k_x, k_y)}} \sin k_y \end{aligned}$$

From second to third line, we used MBZ periodicity. The k_y sum at last line is null: even if using antiperiodic harmonics for $\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$, we did not break inversion symmetry neither in x nor y direction. Bogoliubov bands $E_{\mathbf{k}}$ and thermal factors must show said symmetry. The product of a symmetric function ($\tilde{\text{Th}}/\tilde{E}$) with $\sin k_y$ integrates to zero. This ends the point.

2. $v^{(p_x)} = 0$. To see this, just plug $Q = 0$ in Eq. (3.60),

$$v^{(p_x)} = -V v^{(p_x)} F[s_x^2]$$

Using the positivity rule of Eq. (3.44) we get

$$V > 0 \quad \text{and} \quad F[s_x^2] \geq 0 \quad \implies \quad v^{(p_x)} = 0$$

3. $v^{(p_x)} = 0$. Same proof as point above.

This ends this short section and shows that also for the aAF phase all gap complexifiers actually vanish. This is trivially coherent with the constraint of Eq. (3.40). As a consequence, there is no need to distinguish two AF phases,

$$\text{sAF} = \text{aAF}$$

and gap complexification is a completely “ghost” effect, even if theoretically allowed. All is left is to check for the behaviour of $u^{(d)}$.

1.7.4 Natural suppression of nematicity within $C_2/\mathbb{Z}_2$ symmetry

A consequence of the two sections above is that the gap function is purely real, non-renormalized and constant across all reciprocal space,

$$\tilde{\Delta}_{\mathbf{k}} = mU \quad \implies \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (mU)^2}$$

Bogoliubov bands maintain renormalization through the $u^{(\gamma)}$ HFPs channel. Aim of this section is to prove that, as was for the normal phase, no nematicity occurs in the AF phase, giving $u^{(d)} = 0$.

While the proof given in Sec. 2.5.2 was long and quite elaborated, this one is actually very short. Let us start from recalling the normal phase context:

$$\begin{aligned} \text{Normal phase:} \quad u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

while, using π periodicity, in the present context we can re-write all equations doubling the integration domain,

$$\begin{aligned} \text{AF phase:} \quad u^{(s^*)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\ u^{(d)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \end{aligned}$$

Let us define the pseudo distribution

$$f'(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \equiv -\frac{\tilde{\epsilon}_{\mathbf{k}}}{2\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

which, substituted in the self-consistency equations, makes them very similar to the normal ones, up to a $f \rightarrow f'$ exchange.

The fact that $u^{(d)} = 0$ follows from the simple consideration that in Sec. 2.5.2 (when showing the same for the normal phase) we never used the specific form of f ; instead, we only made use of it being monotonically descendent as a function of $\epsilon_{\mathbf{k}}$. If f' also is a monotonically descendent

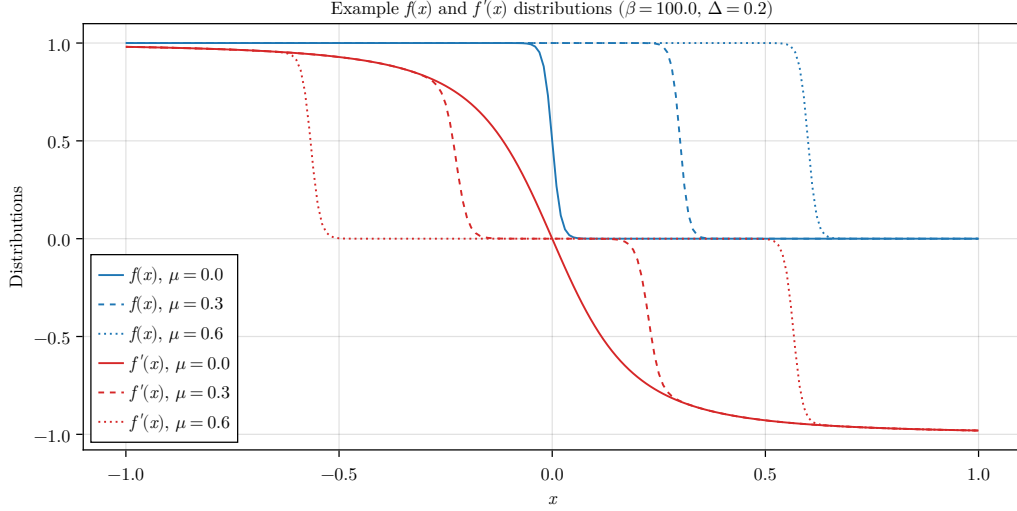


Figure 1.2 | Sketch of the Fermi-Dirac distribution $f(x)$ and the pseudo-distribution $f'(x)$ described in text for the normal phase. These example data have been plotted arbitrarily at $mU = \Delta = 0.2$ for $\mu = 0.0, 0.3, 0.6$.

function of $\epsilon_{\mathbf{k}}$, we can re-use the entire derivation of Sec. 2.5.2. Let us show that using a somewhat lighter notation,

$$\epsilon_{\mathbf{k}} \rightarrow x \quad mU \rightarrow \Delta$$

and omitting parametric arguments. The function f' is given by the product

$$f'(x) = g(x)h(x)$$

with

$$g(x) = -\frac{x}{\sqrt{x^2 + \Delta^2}}$$

$$h(x) = \left(\frac{1}{e^{\beta(\sqrt{x^2 + \Delta^2} - \mu)} + 1} - \frac{1}{e^{\beta(-\sqrt{x^2 + \Delta^2} - \mu)} + 1} \right)$$

The following points are true:

1. The function is antisymmetric about $x = 0$. This is evident by considering that $g(x)$ is evidently antisymmetric, while $h(x)$ is symmetric.
2. $g(x)$ is monotonically descendent. Its derivative is

$$\begin{aligned} \frac{dg}{dx} &= -\frac{\sqrt{x^2 + \Delta^2} - x^2/\sqrt{x^2 + \Delta^2}}{x^2 + \Delta^2} \\ &= -\frac{\Delta^2}{(x^2 + \Delta^2)^{3/2}} \end{aligned}$$

clearly always negative.

3. The product $g(x)h(x)$ is monotonically descendent at positive finite doping. To assert this, we can compute the derivative of f' ,

$$\frac{df'}{dx} = \frac{dg}{dx}h(x) + g(x)\frac{dh}{dx}$$

Now, evidently $h(x) > 0$ and as we said $dg/dx < 0$, which makes the first term negative. The second is negative too: let

$$E(x) \equiv \sqrt{x^2 + \Delta^2} \quad \implies \quad \frac{dE}{dx} = \frac{x}{E(x)} = -g(x)$$

Then:

$$\begin{aligned} \frac{dh}{dx} &= \frac{d}{dx} \left[\frac{1}{e^{\beta(-E(x)-\mu)} + 1} - \frac{1}{e^{\beta(E(x)-\mu)} + 1} \right] \\ &= \beta \frac{dE}{dx} \underbrace{\left[\frac{e^{\beta(-E(x)-\mu)}}{[e^{\beta(-E(x)-\mu)} + 1]^2} + \frac{e^{\beta(E(x)-\mu)}}{[e^{\beta(E(x)-\mu)} + 1]^2} \right]}_{k(x)} \end{aligned}$$

The term in square brackets is obviously positive, $k(x) > 0$. Taking all together, we have

$$g(x) \frac{dh}{dx} = -\beta g^2(x) k(x) < 0 \quad \implies \quad \frac{df'}{dx} < 0$$

making f' descendent monotonicity proven.

These points end the proof: as anticipated, f' is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, a feature that allows us to re-use the entire derivation of Sec. 2.5.2 without hesitation, and conclude that also for the AF phase

$$u^{(d)} = 0$$

while $u^{(s^*)}$ remains constrained in the limits of Eq. (3.41). Starting from a rather complicated scenario, we were able to resolve all the AF HFPs down to the pair

$$m, u^{(s^*)}$$

Perhaps not-so-excitingly, the real net effect of the non-local extension of the HM is, once again, the shrinking effect on bands. Now it is a matter of numeric analysis to assess if Mott localization takes place.

1.8 Discussion of numeric results

In this section we present numeric results. For the AF phase, the HFPs vector, as we thoroughly showed, has two components

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \quad \mathbf{v} = \mathbb{R}^2$$

The resulting model RMPs are, as before:

$$\begin{aligned} \tilde{t}_{ij} &\equiv \left(t - \frac{u^{(s^*)} V}{2} \right) \varphi_{ij}^{(s^*)} \\ \tilde{\mu} &\equiv \mu + n(2zV - U) \end{aligned}$$

In the end, the computational scenario is not so different from the normal phase. The only real difference is our choice of parametric boundaries for simulations:

$$U = 0, 0.05, 0.1, \dots, 8.0 \quad V = 0, 0.05, 0.1, \dots, 8.0 \quad \delta = 0.0$$

We set $\delta = 0.0$. As we anticipated in Sec. 3.5, any finite doping implies immediate disruption of commensurate antiferromagnetism, a consideration that makes any simulation at $\delta > 0$ of low physical significance. Following this consideration, we omit data at finite $\delta \neq 0$.

1.8.1 Magnetization boost

The convergence values for m computed in the UV plane are shown in the panels of Fig. 3.3, along with a level curves plot to identify isomagnetic curves (right panel). Note that, to increase graphic clarity in heatmaps bottom left sector, from the plots are omitted the points at $U \geq 4$ and $V \geq 6$. Tolerances on density and HFPs have been set to

$$\Delta n = 10^{-2} \quad \Delta \mathbf{v} = \begin{bmatrix} m: 5 \times 10^{-4} \\ u^{(s^*)}: 5 \times 10^{-4} \end{bmatrix}$$

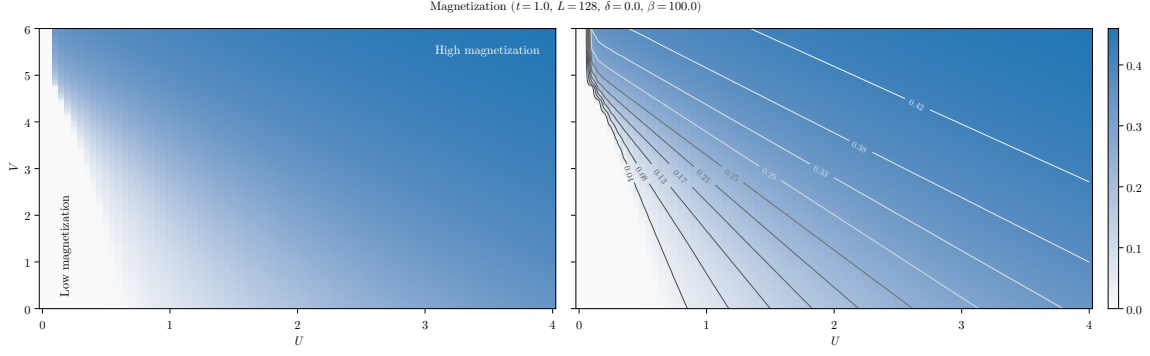
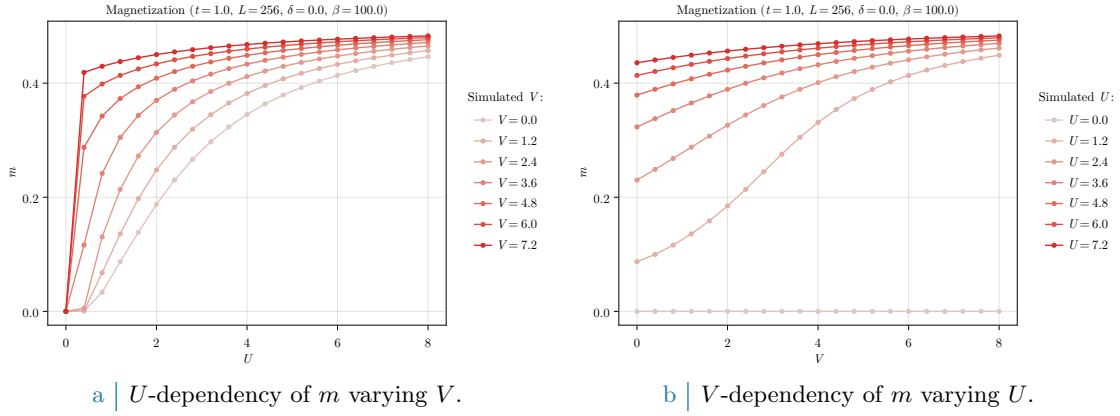


Figure 1.3 | Plot of the self-consistent convergence values for m (left panel) and isomagnetic curves computed on the same data (right panel), along which magnetization is constant. For the sake of graphic clarity, only the points at $0 \leq U \leq 4$ and $0 \leq V \leq 6$ are plotted. [Are $U = 0.05$ data correct?]



a | U -dependency of m varying V .

b | V -dependency of m varying U .

Figure 1.4 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s*)}$ in the AF phase.

The most interesting evident visible directly from heatmaps is that $\hat{H}_V$ actually boosts magnetization. This phenomenon is intuitive and easy to explain: since the only observable net effect of the non-local interaction is to lower t by shrinking bands, at any fixed point (t, U, V) the model can be effectively mapped to

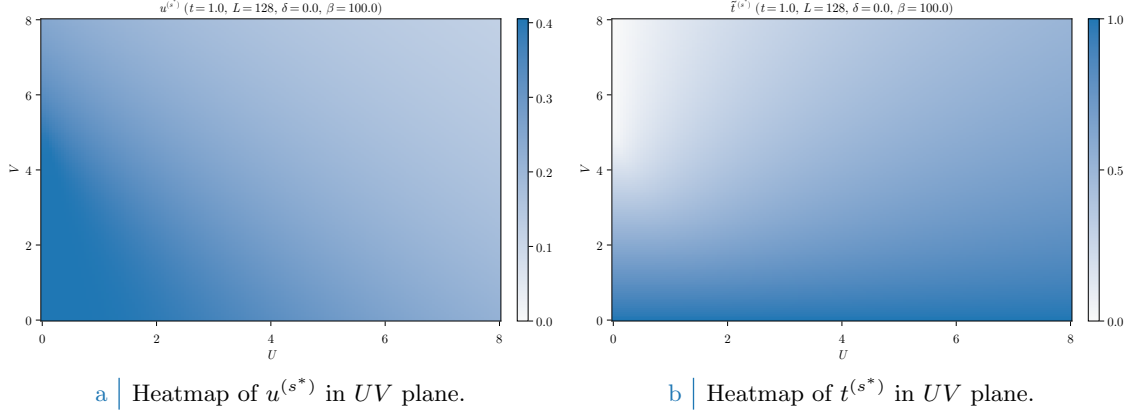
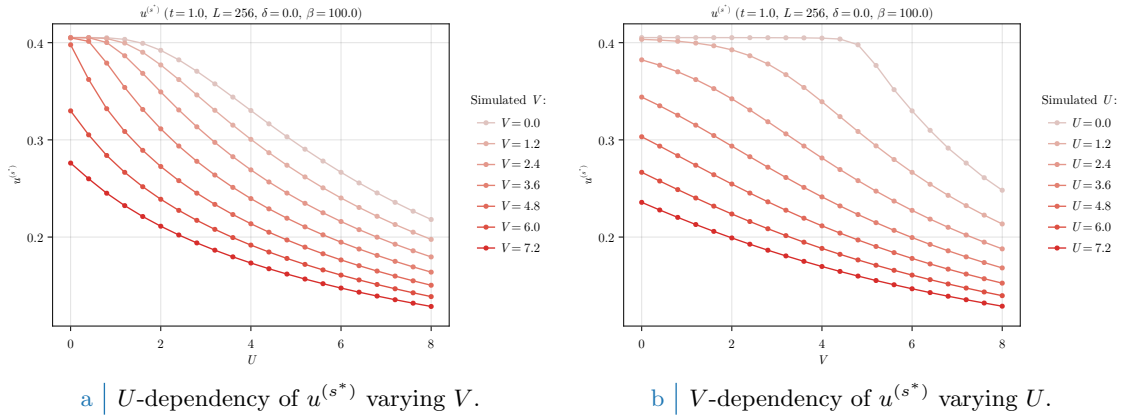
$$\text{EHM: } (t, U, V) \rightarrow \text{HM: } (\tilde{t}, U, 0)$$

being $\tilde{t}$ dependent on V . For the pure HM, to magnetize the system what matters is just U/t . To lower t is physically the same as to increase U , and from this point comes the magnetization precession visible in Fig. 3.3. Isomagnetic lines on the right panel show, for U significantly greater than 0, interesting linear trends whose angular coefficient depend on m itself. The low- U limit of isomagnetic lines presents some oscillations that are evidently imputable to numeric discretization of parametric space. [Add: derivation and discussion about the presence of isomagnetic lines.]

The data shown in Fig. 3.4 confirm clearly the aforementioned magnetization “boost” given by $\hat{H}_V$. The effect is more enhanced at low values of U . Interestingly enough, as is visible from Fig. 3.4b, magnetization does not always saturate the same way. For example, considering the curve at $U = 1.2$, magnetization varies following a somewhat sigmoid shape, making the boost most effective around $V \approx 2 \div 4$. Because of m saturation, this behaviour is lost at higher U s.

1.8.2 Bands shrinking

Bands shrinking effect was discussed in Sec. 3.7.1 within C_4 symmetry, and then confirmed in Sec. 3.7.4 when trying to extend to nematic structures. As was for the normal phase, we have

Figure 1.5 | Data for the HFP $u^{(s*)}$ and $\tilde{t}^{(s*)}$ varying both U and V .Figure 1.6 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s*)}$ in the AF phase.

proven that $\hat{H}_V$ net effect on the hamiltonian kinetic part is to shrink bands,

$$t \rightarrow \tilde{t} < t$$

For the normal phase, in Sec. 2.6.1 we presented numeric results showing a Mott-localized region appears in (δ, V) plane, whose $\delta \rightarrow 0$ limit was approximately $V = 4.93t$, as shown in Fig. 2.7b.

The convergence values for the HFP $u^{(s*)}$ are shown in Fig. 3.5a, and more in details in the two panels of Fig. 3.6. In general, $u^{(s*)}$ decreases monotonically both in U and in V . As we already discussed in normal phase, the $U = 0$ data of Fig. 3.6b present a plateau up to a Mott-critical value. This effect is absent at positive U s. Both interactions disrupt the HFP; however, the two limits

$$U \gg 1 \quad \text{and} \quad V \gg 1$$

are much different. By computing the value of $\tilde{t}^{(s*)}$,

$$\tilde{t}^{(s*)} = 2t - u^{(s*)}V$$

we obtain the data of Fig. 3.5b. The limit $U \gg 1$ taken at fixed V , also shown in Fig. 3.7a, provides “ U -induced antiferromagnetism”, with bands shrinking effect being almost irrelevant as V lowers; on the contrary, the opposite limit $V \gg 1$ taken at fixed U , also shown in Fig. 3.7b, shrinks bands strongly providing a “Mott-induced antiferromagnetism”.

Interestingly, in (U, V) space there is not Mott-localized region of the kind we had in (δ, V) space for the normal phase (see Fig. 2.7b). At half filling, complete localization only occurs at $U = 0$, as can be seen in the bottom left sector of Fig. 3.7a. Any finite value of U disrupts bands flattening, thus – being the local repulsion an essential ingredient in Hubbard antiferromagnetism

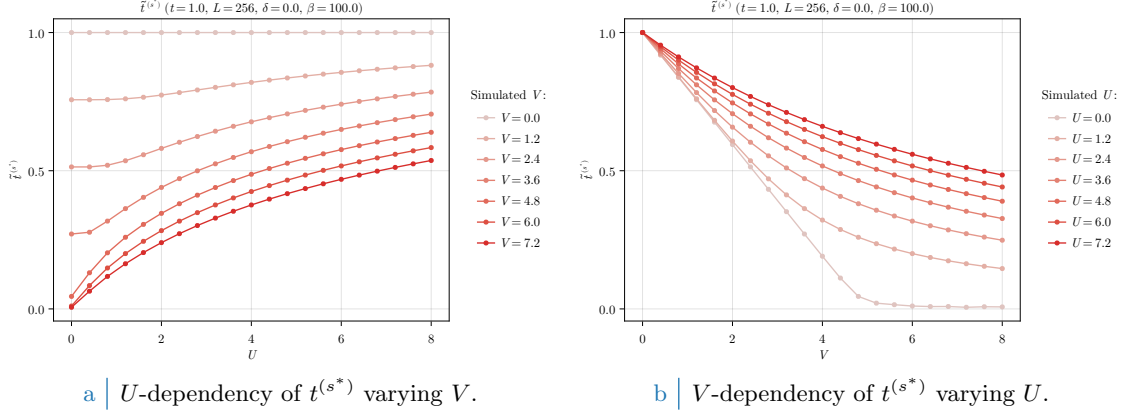


Figure 1.7 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

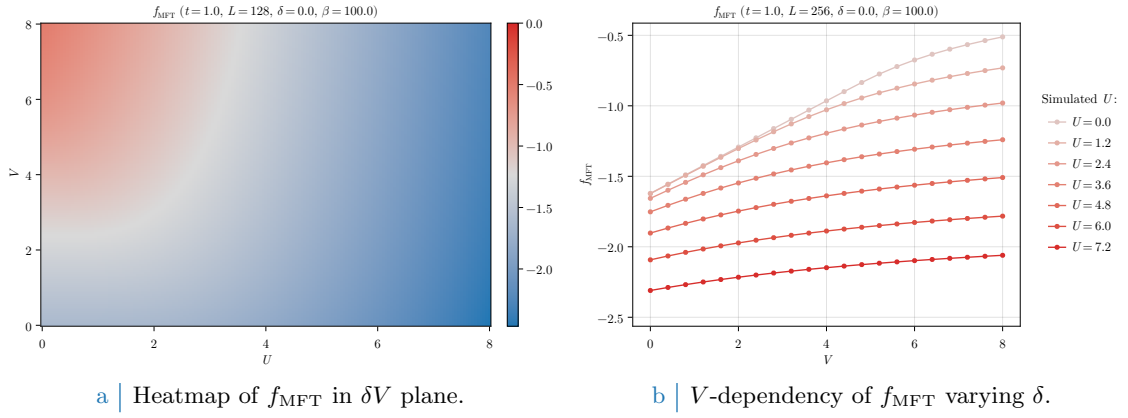


Figure 1.8 | Free energy density of the AF phase.

– we can safely sustain, based on numeric evidence, that bands flattening is incompatible with a commensurate AF phase on the half-filled EHM.

1.8.3 Solution free energy and entropy densities

In Fig. 3.8 the solution free energy is plotted. In the Fig. 3.8b it becomes evident how the energetically most stable state is realized at low V , high U . The opposite limit of weak local repulsion, strong non-local attraction (top-left corner of Fig. 3.8a) instead increases f_{MFT} a lot, correctly reaching zero energy at $U = 0$, $V > 4.93t$ – the critical value discussed in Sec. 2.6.1 and computed in Eq. (2.28).

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 3.3, right panel, and thus the realized state exhibits an identical value of m , much different is the energetic content of the state itself. While in the $U \gg 1$, fixed V limit free energy gain comes from gap amplification – which moves down the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ –, in the $V \gg 1$, fixed U limit energies of the lower Bogoliubov band (full at low temperature) are shifted upwards via the Mott localization effect. [To be continued...]

1.8.4 Fixed-point interpretation of HF solution

As a final remark discussing the mathematical structure of the algorithm, we anticipated in Sec. 1.4.4 how the reject-mix-iterate scheme upon which our HF algorithm is built can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$A(\mathbf{v}) = \mathbf{v}$$

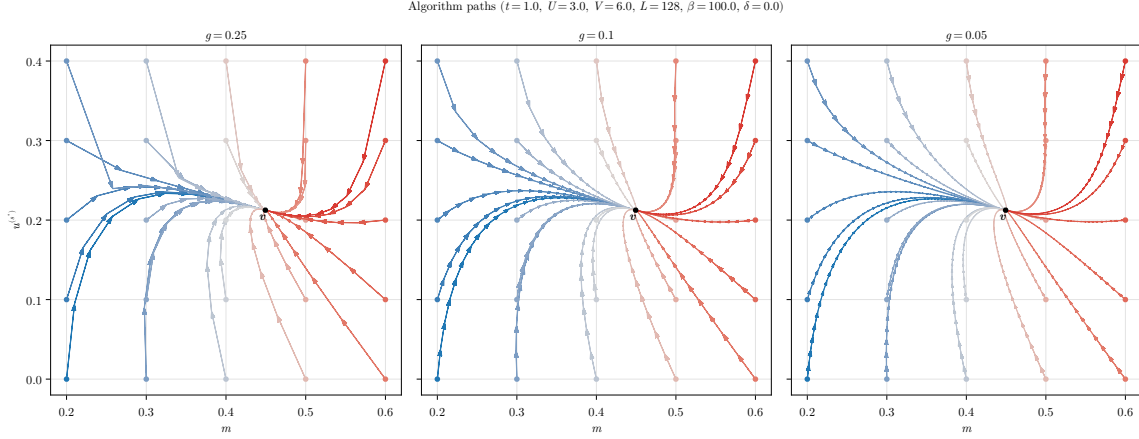


Figure 1.9 | Quiver plot of algorithmic evolution in HFPs space at fixed model parametrization. The black dot $\mathbf{v}$ represents the nonlinear map fixed point, while the colored paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initializers (colored dots).

To predict for a given initializer $\mathbf{v}_0$ the path it will follow through HFPs space is, in general, highly non-trivial due to the intrinsic nonlinearity of the algorithmic map defined by self-consistency equations for the various HFPs. An example of such nonlinearity is given in the derivation of Secs. 3.7.2, 3.7.3.

In Fig. 3.9 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3 \quad V = 6 \quad L = 128 \quad \beta = 100 \quad \delta = 0$$

At this point, the self-consistent HF “fixed-point” solution is given by

$$m \approx 0.449 \quad u^{(s^*)} \approx 0.212$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initializers in the set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} : \quad m_0 \in \{0.2, 0.3, \dots, 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0, 0.1, \dots, 0.4\}$$

In figures, different colors indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one step to the following.

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g . To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrization could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map “softer” and curvier, as is in the right panel of Fig. 3.9. It’s not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

Chapter 2

Mean-Field Theory in Hubbard lattices

In this Appendix the MFT solutions to the conventional Hubbard hamiltonian of Eq. (??),

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad t, U > 0 \quad (2.1)$$

are described. The discussion is limited to the two-dimensional square lattice. The results of this appendix are relevant and used throughout the text as the groundwork to be extended by the means of MFT applied to the non-local part of the EHM.

2.1 Antiferromagnetic phase

We work in a commensurate AF phase ordering, which is, we explicitly break symmetry by realizing a SDW phase with spin density unevenly distributed on sites with periodicity of two sites, each direction. Recalling the discussion of Sec. 1.1, AF long-range ordering we deal with here is described by the composite SSB

$$\begin{aligned} \text{SU}(2) &\xrightarrow{\text{SSB}} \text{U}^z(1) \\ \text{T}_1^{(x)} \otimes \text{T}_1^{(y)} &\xrightarrow{\text{SSB}} \text{T}_2^{(x)} \otimes \text{T}_2^{(y)} \end{aligned}$$

Recalling Tab. 1.3, for such a physical phase the only Wick contraction channel available is the Hartree one. Let us proceed in such sense. As in Eqns. (3.1), (3.2), it is useful to define the density and AF fluctuation operators:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \quad (2.2)$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \quad (2.3)$$

whose EVs are relevant order parameters for the normal and AF phases.

2.1.1 MFT treatment of the Hartree channel

In terms of the physical magnetization m , the MFT Ansatz is given by

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \quad (2.4)$$

Decoupling the HM hamiltonian and substituting,

$$\begin{aligned} \hat{H} &\simeq -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle] \\ &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] - mU \sum_{\mathbf{r}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] - E_{\text{H}/U}^{(\text{AF})} \quad (2.5) \end{aligned}$$

with $E_{H/U}^{(\text{AF})}$ the HM Hartree contraction energy,

$$\begin{aligned} E_{H/U}^{(\text{AF})} &\equiv U \sum_{\mathbf{r}} \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= U \sum_{\mathbf{r}} (n + (-1)^{x+y} m) (n - (-1)^{x+y} m) \\ &= L_x L_y \times U (n^2 - m^2) \end{aligned} \quad (2.6)$$

By a Fourier transform, the phase factor of Eq. (??) can be absorbed in the destruction operator inside of $\hat{n}_{\mathbf{r}\sigma}$:

$$\begin{aligned} \sum_{\mathbf{r}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}} (-1)^{x+y} \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \\ &\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r}} e^{i\boldsymbol{\pi} \cdot \mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\ &= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\mathbf{k}' \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}} e^{-i[\mathbf{k}' - (\mathbf{k} + \boldsymbol{\pi})] \cdot \mathbf{r}} \\ &= \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma} \end{aligned} \quad (2.7)$$

where $\boldsymbol{\pi} = (\pi, \pi)$. It follows:

$$mU \sum_{\mathbf{r}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] = mU \sum_{\mathbf{k} \in \text{BZ}} [\hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\downarrow}]$$

Note that the Hartree channel renormalizes chemical potential,

$$nU \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] = nU \times \hat{N} \implies \mu \rightarrow \tilde{\mu} \equiv \mu - nU$$

an effect that is present also in the EHM. The kinetic part of the hamiltonian transforms as in Eq. (1.13),

$$-t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}] \stackrel{\mathcal{F}}{=} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma}$$

being $\epsilon_{\mathbf{k}}$ the bare bands of Eq. (1.14).

Take everything together. Using the property:

$$\begin{aligned} \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k} \in \text{MBZ}} [\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma} + \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma}^\dagger \hat{c}_{\mathbf{k} + 2\boldsymbol{\pi}\sigma}] \\ &= \sum_{\mathbf{k} \in \text{MBZ}} [\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma} + \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma}] \end{aligned} \quad (2.8)$$

and reduce Eq. (??) to its reciprocal form

$$\hat{H} \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} [\hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k} + \boldsymbol{\pi}\sigma}] - mU \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} [\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger] + nU \hat{N} - E_{H/U}^{(\text{AF})}$$

2.1.2 More on AF Ansatz

Before moving on, let us give a little detail on the AF Ansatz. The decoupling in Hartree sector of the density-density interaction can be treated with more care: denoting by d.c. the double counting terms we ignore,

$$\begin{aligned} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} - \text{d.c.} &= \hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} \\ &= \hat{n}_{\mathbf{r}\uparrow} \frac{\langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} + \hat{n}_{\mathbf{r}\uparrow} \frac{\langle \hat{n}_{\mathbf{r}\downarrow} \rangle - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} + \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} \hat{n}_{\mathbf{r}\downarrow} + \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} \hat{n}_{\mathbf{r}\downarrow} \\ &= \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}) - \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}) \end{aligned}$$

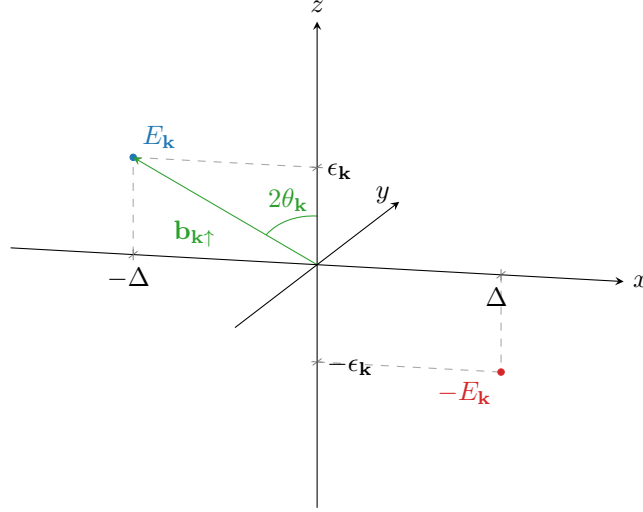


Figure 2.1 | Pseudo-magnetic field originating from mean-field treatment of the square Hubbard hamiltonian. Here, only the $\sigma = \uparrow$ situation is plotted.

Our commensurate AF Ansatz essentially states that each site has the same spin-averaged density,

$$n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad \text{is } \mathbf{r}\text{-independent}$$

while the spin averaged unbalance has a two-sites periodicity

$$m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad \text{is } \mathbf{r}\text{-independent}$$

This way we reduce translational invariance. Of course, if the two quantities above are $\mathbf{r}$ -independent indeed, it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (2.9)$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (2.10)$$

While from the first equation we gain an operational method to compute density, from the second we get a magnetization self-consistency equation.

2.1.3 Nambu representation of the AF phase and diagonalization

We now map the reciprocal version of the HM onto a simpler problem. Throughout this section we use the notation $\epsilon_{\mathbf{k}} \rightarrow \epsilon_{\mathbf{k}\sigma}$ with identical meaning. Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} m U$$

$$\hat{H} \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^\dagger h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E_{\text{H}/U}^{(\text{AF})} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma}^* \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \quad (2.11)$$

Notice: the Nambu hamiltonian is a 2×2 matrix over the MBZ – which is half the full BZ, coherently with a solution which essentially bipartites the lattice giving back a double sized unit cell.

The system ground-state is obtained by the means of a Bogoliubov rotation. The hamiltonian maps onto the simple one of a spin in a magnetic field (the gap is purely real here),

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} \tau^z - \Delta_{\mathbf{k}\sigma} \tau^x$$

being τ^α the Pauli matrices. Then, defining

$$\hat{s}_{\mathbf{k}\sigma}^\alpha \equiv \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^\alpha \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{and} \quad \mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

one gets:

$$\hat{H} \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} - E_{\text{H/U}}^{(\text{AF})} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}_{\mathbf{k}\sigma}^x \\ \hat{s}_{\mathbf{k}\sigma}^y \\ \hat{s}_{\mathbf{k}\sigma}^z \end{bmatrix} \quad (2.12)$$

The hamiltonian represents a system of spins subject to local magnetic fields, each tilted by an angle $\tan 2\theta_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma}/\epsilon_{\mathbf{k}\sigma}$, as sketched in Fig. ???. Evidently the following relations, also reported in Tab. 3.2, hold:

$$\hat{s}_{\mathbf{k}\sigma}^x = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \quad (2.13)$$

$$\hat{s}_{\mathbf{k}\sigma}^y = i \left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) \quad (2.14)$$

$$\hat{s}_{\mathbf{k}\sigma}^z = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\pi\sigma} \quad (2.15)$$

Let also:

$$\begin{aligned} \hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \mathbb{I}_{2 \times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \end{aligned} \quad (2.16)$$

Diagonalization of each $h_{\mathbf{k}\sigma}$ is obtained trivially by a simple rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^\dagger \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix}$$

and where the diagonalization unitary matrix is simply a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, counter-clockwise

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma} \tau^y} \\ &= \cos \theta_{\mathbf{k}\sigma} - i \sin \theta_{\mathbf{k}\sigma} \tau^y \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} & -\sin \theta_{\mathbf{k}\sigma} \\ \sin \theta_{\mathbf{k}\sigma} & \cos \theta_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \quad (2.17)$$

Eigenvalues are:

$$E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \quad (2.18)$$

At any finite temperature, the ground-state of such a system is well-known: a thermal superposition of “down” spins with “up” spins. In the pseudospin picture, at each “site” ($\mathbf{k}\sigma$), the lowest-energy state is the one anti-aligned with the pseudo-magnetic field. It is useful now to define the thermal factors,

$$\text{Th}_{\mathbf{k}\sigma}^{(\pm)}[\beta, \mu] \equiv f(-E_{\mathbf{k}\sigma}; \beta, \mu) \pm f(E_{\mathbf{k}\sigma}; \beta, \mu) \quad (2.19)$$

that will appear quite often. Then it's immediate to derive the various expectation values.

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = \frac{\text{Re}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (2.20)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle = \frac{\text{Im}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (2.21)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (2.22)$$

Note that considering the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and using Eq. (??) the diagonalizing matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k} \in \text{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \implies W_{\mathbf{k}\sigma} [h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2}] W_{\mathbf{k}\sigma}^\dagger = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2} \quad (2.23)$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. In other words, the new bands retain the same chemical potential of the bare bands. Notice also that the presence of an absolute value makes the eigenvalues independent of the σ index. Eigenvector fermion operators are obtained simply as:

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} \quad (2.24)$$

The $\hat{\Gamma}$ spinor operators are effectively free fermionic spinor fields and as such must be treated. Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}\sigma} &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} & -\sin \theta_{\mathbf{k}\sigma} \\ \sin \theta_{\mathbf{k}\sigma} & \cos \theta_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma} \\ \sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} + \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix} \end{aligned} \quad (2.25)$$

The consequence of Eq. (??) is interesting by itself: within this HF solution, the gap opens always at $E = 0$. In other words, the bands sketched in Fig. ?? retain the same geometric features regardless of the doping, what changes is the bands width and the gap between them. The very key point here is that, being the two bands symmetric (thus containing each half of the electronic states) and defined over the MBZ, any finite doping $\delta > 0$ will produce a metallic behaviour, imposing that the chemical potential crosses the upper band. It is widely observed and confirmed by more refined numerical procedures, however, that AF establishes an insulating phase – here only reproduced at half-filling, where the chemical potential lies within the gap. As will be detailed in ??, the commensurate nature of AF SSB at wavevector $\boldsymbol{\pi}$ of this self-consistent solution is highly unstable and atypical at finite doping [15].

2.1.4 Explicit zero-temperature AF ground state at half-filling

The AF ground state can be expressed easily at $n = 1/2$ and $\beta \rightarrow +\infty$. Let $|\uparrow_{\mathbf{k}\sigma}\rangle$ and $|\downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the “up” and “down” $\mathbf{k}$ -th pseudospin states. Let $|\overline{\downarrow}_{\mathbf{k}\sigma}\rangle$ be the “down” state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle \quad (2.26)$$

We can link the “down” states via an inverse rotation,

$$\begin{aligned} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle &= e^{-i\theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y} |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - i \sin \theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} (\hat{s}_{\mathbf{k}\sigma}^+ - \hat{s}_{\mathbf{k}\sigma}^-) |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} |\uparrow_{\mathbf{k}\sigma}\rangle \end{aligned} \quad (2.27)$$

having we used

$$\hat{s}_{\mathbf{k}\sigma}^{\pm} \equiv \frac{\hat{s}_{\mathbf{k}\sigma}^x \pm i\hat{s}_{\mathbf{k}\sigma}^y}{2}$$

Use now Eqns. (??) and (??),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}\sigma} \quad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}+\pi\sigma}$$

Evaluating the above operators on the “up” and “down” states, we get immediately

$$\begin{aligned} \frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 1 & \frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 0 \\ \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 0 & \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 1 \end{aligned}$$

Thus, up to a phase

$$|\uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}\sigma}^\dagger |\Omega_{\mathbf{k}\sigma}\rangle \quad |\downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger |\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k}, \sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (??) and (??) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger + e^{2i\zeta_{\mathbf{k}\sigma}} \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (2.28)$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (??). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned} \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle &= (e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}) \sin \theta_{\mathbf{k}\sigma} \cos \theta_{\mathbf{k}\sigma} \\ &= \cos 2\zeta_{\mathbf{k}\sigma} \sin 2\theta_{\mathbf{k}\sigma} \end{aligned}$$

but from Eq. (??)

$$\begin{aligned} \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle &= \langle \overline{\downarrow}_{\mathbf{k}\sigma} | \hat{s}_{\mathbf{k}\sigma}^x | \overline{\downarrow}_{\mathbf{k}\sigma} \rangle \\ &= \sin 2\theta_{\mathbf{k}\sigma} \end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \implies \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with } k \in \mathbb{Z}$$

and lets us conclude from Eq. (??)

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger + \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (2.29)$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system. Finally, using Eq. (??), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \gamma_{\mathbf{k}\sigma}^{(-)\dagger} |\Omega_{\mathbf{k}\sigma}\rangle \quad (2.30)$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a π shift in reciprocal space, producing a new free gas where all interactions have been “condensed” in these AF pairs.

2.1.5 Spin degeneracy removal

Up to now, we wrote explicitly the σ index. However,

$$\epsilon_{\mathbf{k}\uparrow} = \epsilon_{\mathbf{k}\downarrow} \quad \Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$$

thus $E_{\mathbf{k}\uparrow} = E_{\mathbf{k}\downarrow}$. Since the gap function sign does not have a measurable impact, we might as well take the spin index as a pure degeneracy and from now on work in the $\sigma = \uparrow$ sector, dropping the σ index. We can further simplify by dropping all indices from the gap function,

$$\Delta_{\mathbf{k}\uparrow} = mU \equiv \Delta$$

being the latter purely s -wave. This is a property of the pure HM: for the EHM, the gap acquires a non-trivial topological structure.

2.1.6 Chemical potential and system density

Using the fact that

$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \hat{\Psi}_{\mathbf{k}\sigma} \rangle$$

due to the $W_{\mathbf{k}\sigma}$ unitarity properties, the total number of electrons is given by (also recalling Eq. (??))

$$\begin{aligned} N &= \sum_{\mathbf{k}\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \rangle \\ &= \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ &= 2 \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}}^{(+)}[\beta, \mu] \end{aligned} \quad (2.31)$$

being f be the Fermi-Dirac distribution at inverse temperature β and chemical potential μ ,

$$f(\epsilon; \beta, \mu) = \frac{1}{e^{\beta(\epsilon - \mu)} + 1}$$

The 2 prefactor appeared because of spin degeneracy. Since the gapped bands refer to the smaller MBZ, thus exhibit the periodicity

$$E_{\mathbf{k}} = E_{\mathbf{k}+\pi}$$

it holds equivalently

$$N = \sum_{\mathbf{k} \in \text{BZ}} \text{Th}_{\mathbf{k}}^{(+)}[\beta, \mu] \quad (2.32)$$

as is obvious. The 2 prefactor was absorbed in doubling the integration region, $\text{MBZ} \rightarrow \text{BZ}$. Next sections are devoted to simplify the above theoretical results in order to obtain an algorithmic estimation for the magnetization m , which is just the AF instability order parameter.

In Fig. ?? the density $n(\mu)$ is plotted as a function of μ . As is evident, at nearly zero temperature ($\beta \rightarrow \infty$) the system is half filled for $-\Delta < \mu < \Delta$. In order to obtain a finite doping, it must be $|\mu| > \Delta$; which means, the chemical potential needs to cross the electronic bands. This HF solution leads invariably to a rather unusual metallic AF at any finite doping.

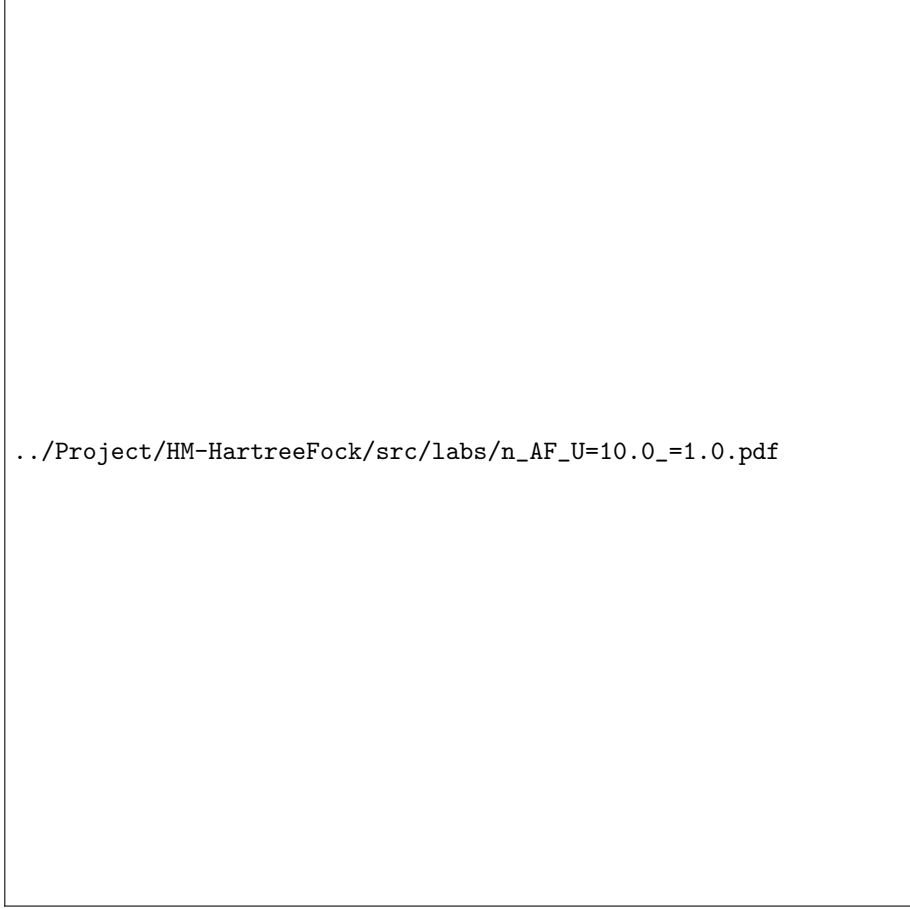
2.1.7 Self-consistency gap equation and magnetization

A convergence algorithm can be designed to find the Hartree-Fock solution to the model. Ultimately, we aim to extract m self-consistently. Combining Eqns. (??), (??) and the definition of Eq. (??), we get

$$\frac{1}{2L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger]$$

thus employing Eq. (??)

$$\begin{aligned} m &= \frac{1}{2L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \langle (\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}_{\mathbf{k}\uparrow}^\dagger) - (\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}_{\mathbf{k}\downarrow}^\dagger) \rangle \end{aligned}$$



../Project/HM-HartreeFock/src/labs/n_AF_U=10.0_1.0.pdf

Figure 2.2 | Electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures.

Using now Eq. (??) we get the self-consistency equation for the HFP m ,

$$\begin{aligned}
 m &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [\langle \hat{s}_{\mathbf{k}\uparrow}^x \rangle - \langle \hat{s}_{\mathbf{k}\downarrow}^x \rangle] \\
 &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\Delta_{\mathbf{k}\uparrow}\} - \text{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu] \\
 &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\Delta}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu]
 \end{aligned} \tag{2.33}$$

The magnetization m is the physical order parameter of the phase transition from the normal phase to the antiferromagnetic one. Within such phase transition, in fact, it is the expectation value

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} \rangle$$

that goes from zero (normal phase, free Fermi gas) to a non-zero value (AF). As is evident from equations above, m is just its (only) s -wave component.

Consider specifically the half-filled situation and multiply Eq. (??) on both sides by U . Since due to symmetry $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\Delta}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right)$$

where we used the relation

$$\text{Th}_{\mathbf{k}}^{(-)}[\beta, 0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$$

In thermodynamic limit $L \rightarrow \infty$

$$\frac{1}{U} \simeq \int_{\text{MBZ}} \frac{d\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2} \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right)$$

Substitute the momentum integral with an integral over energies, in the low temperature limit, $\beta \rightarrow \infty$ (thus approximate the hyperbolic tangent with 1)

$$\frac{1}{U} \simeq \int_{\text{bands}} \frac{d\epsilon \rho(\epsilon)}{\sqrt{\epsilon^2 + \Delta^2}}$$

[To be continued...]

2.1.8 Instability of the commensurate AF solution and atypical metallic bands

A key point to be discussed is the known instability of the AF HF solution of the EHM. As is thoroughly discussed in [15], the solution we presented above is only stable at half-filling, becoming hardly unstable at infinitesimal doping. This is due to the commensurate nature of the established AF phase: we impose SSB to take place at wavevector $\boldsymbol{\pi}$, shrinking the BZ to exactly half its size and giving back on the newly defined MBZ two bands $\pm E_{\mathbf{k}}$. This procedure actually works well for the half-filled case: the magnetization essentially depends on the scattering rate between points on the FS connected by a $\boldsymbol{\pi}$ vector. The half filled case, taking advantage of the natural nesting property of the bare bands,

$$\forall \mathbf{k} \in \partial\text{MBZ} : \epsilon_{\mathbf{k}} = \epsilon_{\mathbf{k}+\boldsymbol{\pi}}$$

(being ∂MBZ the boundary of the MBZ) exhibits a magnetization at infinitesimal interaction. For a many body state resulting from continuous SSB, to establish a given phase means to show the presence of a stable gapless Goldstone mode associated to the broken symmetry. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the AF phase, it follows from the reduced $U^z(1)$ rotational symmetry and is related to the proper transverse pseudospin-pseudospin response function, 2×2 matrix equation,

$$\tilde{\chi}_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\mathbf{k}, \omega) = \frac{\chi_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\mathbf{k}, \omega)}{1 - U \chi_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\mathbf{k}, \omega)} \quad (2.34)$$

being χ the free transverse response. The pseudospin hereby considered is the one of Eq. (??). As is shown in [15], for the commensurate AF SSB at wavevector $\boldsymbol{\pi}$ the condition for the presence of a stable gapless mode reduces to a 2×2 matrix equation,

$$\text{Stability condition:} \quad \mathbb{I}_{2 \times 2} - U [\chi_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\boldsymbol{\pi}, 0)] \geq 0 \quad (2.35)$$

with $[\chi_{\hat{\mathbf{s}}-\hat{\mathbf{s}}+}(\boldsymbol{\pi}, \omega)]$ the response function matrix obtained by considering a matrix response function for a two-sites unit cell. The above equation is to be interpreted as a matricial equation [Explain better..]. Considering the eigenvalue of largest amplitude of the matrix λ_{δ} at filling δ , the one most prone to instabilities, it is found that

$$\begin{aligned} \lambda_0(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) &= \frac{1}{U} & (\text{half filling}) \\ \lambda_{\delta}(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) &= \frac{1}{U} + \frac{2\epsilon_F}{\Delta} N(\epsilon_F) & (\text{finite doping}) \end{aligned}$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. At half-filling the system is an insulator, therefore no Fermi energy is defined, as well as no FS. Evidently this implies

$$\begin{aligned} 1 - U \lambda_0(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) &= 0 & (\text{half filling}) & \implies \text{stable Goldstone mode} \\ 1 - U \lambda_{\delta}(\mathbf{k} \rightarrow \boldsymbol{\pi}, 0) &< 0 & (\text{finite doping}) & \implies \text{unstable Goldstone mode} \end{aligned}$$

For any finite doping, the commensurate AF phase becomes unstable and is destroyed by transverse spin fluctuations. Nevertheless, for the sake of completeness and because we noticed too late, we will study the metallic AF also at finite doping.

2.1.9 Hartree-Fock algorithm results in reciprocal space

The above sections offer a self-consistency equation for the only HFP m ; W is determined by $\Delta = mU$ indeed. In an iterative HF fashion, Eq. (??) reads:

$$m_{i+1} = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{m_i}{2E_{\mathbf{k}}[\Delta_i]} [f(-E_{\mathbf{k}}[m_i]; \beta, \mu) - f(E_{\mathbf{k}}[m_i]; \beta, \mu)] \quad (2.36)$$

Then a self-consistent algorithm to search for a self consistent estimate of the magnetization may be sketched, following the general idea of Sec. 1.4:

1. Algorithm setup: initialize a counter $i = 1$ and choose:

- the local repulsion U/t or equivalently the hopping amplitude t/U ;
- the coarse-graining of the BZ, which is, fix L_x and L_y . Then

$$k_x = 2\pi n_x / L_x \quad k_y = 2\pi n_y / L_y$$

where $-L_j/2 \leq n_j \leq L_j/2$, $n_j \in \mathbb{Z}$;

- the density n or equivalently the doping $\delta \equiv n - 0.5$;
- the inverse temperature β ;
- the maximum number of iterations $p \in \mathbb{N}$;
- the mixing parameter $g \in [0, 1]$;
- the tolerance parameters $\delta_m, \delta_n \in \mathbb{R}$ (respectively for the HFP m and density);

2. Select a random starting value $m_i > 0$ for $i = 1$;

3. Find the optimal chemical potential μ as follows:

- a) Compute $\pm E_{\mathbf{k}}[\Delta_i, U]$;
- b) Define the function of the chemical potential μ ,

$$\delta n(\mu; \beta, m_i, U) \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [f(-E_{\mathbf{k}}[U, m_i]; \beta, \mu) + f(E_{\mathbf{k}}[U, m_i]; \beta, \mu)] - n$$

c) Find the root of δn ,

$$\delta n(\mu_i) = 0$$

Then μ_i is the chemical potential which, at step i with HFP m_i , realizes the best approximation of density n ;

4. Compute m_{i+1} using Eq. (??) with chemical potential μ_i and update the counter, $i \rightarrow i + 1$;

5. Check if $|m_{i+1} - m_i| \leq \delta_m$:

- If yes, halt;
- If not: check if $i > p$
 - If yes, halt and consider choosing better tolerance and model parameters;
 - If not, define

$$m_{i+1} : gm_{i+1} + (1 - g)m_i$$

(logical assignment notation used) and repeat from step 2.

The algorithm sketched hereby was ran with the following setup:



Figure 2.3 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. ??) and the doping $\delta = n - 1/2$ (Fig. ??).

```

1 UU = [U for U in 0.5:0.5:20.0]      # Local repulsions
2 LL = [256]                          # Lattice sizes
3 dd = [d for d in 0.0:0.05:0.45]    # Dopings
4 bb = [100.0, Inf]                  # Inverse temperatures
5 p = 100                             # Max number of iterations
6 dm = 1e-4                          # Tolerance on magnetization
7 dn = 1e-2                          # Tolerance on density
8 g = 0.5                            # Mixing parameter

```

In Fig. ?? plots at $\beta = 100$ of the magnetization m is reported. As is evident in Fig. ??, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger, a mere consequence of the fact that the free space at each single-particle state in k -space shrinks as density increases; also, for $\delta > 0$, a finite magnetization m exists only for $U \geq U_c[\delta]$ (a critical value parametrically dependent on doping). Fig. ?? shows the dependence of magnetization on doping at various fixed local repulsions U : once again, to dope the material disrupt AF ordering. Identical analysis have been performed for different temperatures, leading to analogous results as long as $\beta \gtrsim 10$ and to $m \simeq 0$ for higher temperatures (predictably).

2.2 (Impossibility of a stable) superconducting phase

Let us now move to the discussion of superconductivity in the HM. As for the AF phase, we use the results and analyses of this section as groundwork for the more sophisticated case of SC phase in the EHM. We anticipate here that BCS-like superconductivity cannot develop in the pure HM, due to the inherently repulsive nature of $\hat{H}_U$. BCS theory admits, for an half filled model, finite Cooper fluctuations for an infinitesimal attractive interaction. However, as will be discussed, by the means of MFT it is perfectly possible and coherent to derive a self-consistent superconducting solution to the HM. The point here is that such self-consistent solution sets to zero Cooper fluctuations.

As done for the AF phase in Sec. ?? and following the introductory discussion of Sec. [Missing], let us start from the discussion of the phase symmetries. We aim to describe a uniform-density superconductor,

$$\langle \hat{n}_{i\sigma} \rangle = n$$

taking into account Cooper RWCs,

$$\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\bar{\sigma}}^\dagger \rangle$$

which, having we broken $U^c(1)$ charge symmetry, we allow to fluctuate to non-zero values. For the pure HM, we know from Tab. 1.3 that both Hartree and Cooper pairing channels contribute to the RWCs set. We handle them separately.

2.2.1 MFT treatment of the Hartree channel

For the Hartree channel, knowing the deal with a uniform-density phase, we can re-use the derivation of Sec. ?? simply by setting $m = 0$. We immediately get from Eq. (??):

$$\begin{aligned} \hat{H}_U &\simeq U \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle] + (\text{Cooper channel}) \\ &= nU \times \hat{N} - E_{\text{H}/U}^{(\text{SC})} + (\text{Cooper channel}) \end{aligned} \quad (2.37)$$

thus chemical potential renormalization $\tilde{\mu} \equiv \mu - nU$ is present also for the SC phase. The contraction energy comes immediately from Eq. (??),

$$E_{\text{H}/U}^{(\text{SC})} = L_x L_y \times n^2 U \quad (2.38)$$

We now move to Cooper channel.

2.2.2 MFT treatment of the Cooper channel

Recall the result of Eq. (1.21) which gave the Fourier-transformed version of the local repulsion, and perform a Cooper class Wick's contraction on it,

$$\begin{aligned} \hat{H}_U &\simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \left[\langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \right. \\ &\quad \left. + \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \right] \end{aligned} \quad (2.39)$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define as in Eq. (??) the pairing operator

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \quad (2.40)$$

and define $w_{\mathbf{q}}$ as in Eq. (??),

$$w_{\mathbf{q}} \equiv \langle \phi_{\mathbf{q}}^\dagger \rangle$$

Its s -wave part is given by:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} w_{\mathbf{q}} \quad (2.41)$$

Let $\hat{n}_{\mathbf{k}\sigma}$ as in Eq. (??). Eq. (??) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \left[\langle \hat{\phi}_{\mathbf{q}}^\dagger \rangle \hat{\phi}_{\mathbf{q}'} + \hat{\phi}_{\mathbf{q}}^\dagger \langle \hat{\phi}_{\mathbf{q}'} \rangle \right] - E_{\text{C}/U}^{(\text{SC})} \quad (2.42)$$

being $E_{\text{C}/U}^{(\text{SC})}$ the HM Cooper contraction energy,

$$\begin{aligned} E_{\text{C}/U}^{(\text{SC})} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \langle \hat{\phi}_{\mathbf{q}}^\dagger \rangle \langle \hat{\phi}_{\mathbf{q}'} \rangle \\ &= L_x L_y \times U |w^{(s)}|^2 \end{aligned} \quad (2.43)$$

Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

We finally reduce $\hat{H}_U$ to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{q} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{q}}^\dagger \right] - E_{C/U}^{(\text{SC})} \quad (2.44)$$

which can be easily mapped on a pseudospin model.

Composing Eqns. (??) and (??), we get the full MFT hamiltonian

$$\begin{aligned} \hat{H} \simeq \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} - \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} \right] \\ - \sum_{\mathbf{q} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{q}}^\dagger \right] + nU \times \hat{N} - \left[E_{H/U}^{(\text{SC})} + E_{C/U}^{(\text{SC})} \right] \end{aligned} \quad (2.45)$$

2.2.3 Nambu representation of the SC phase and diagonalization

We now proceed precisely as in Sec. ??. Define the SC Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix}$$

Define the pseudospin operator:

$$\hat{s}_{\mathbf{k}}^\alpha \equiv \hat{\Phi}_{\mathbf{k}}^\dagger \tau^\alpha \hat{\Phi}_{\mathbf{k}}$$

and the shifted bare bands,

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the chemical potential RMP. We employ from the start the spin-independency of the bare bands, as well as their pure s^* -wave structure, since no Fock interaction in the present context can change that. The following chain holds:

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow}^\dagger + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{\Phi}_{\mathbf{k}}^\dagger \tau^z \hat{\Phi}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{s}_{\mathbf{k}}^z + \hat{\mathbb{I}}_{\mathbf{k}} \end{aligned} \quad (2.46)$$

Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned} \hat{H}_t - \tilde{\mu} \hat{N} &= \sum_{\sigma} \sum_{\mathbf{k} \in \text{BZ}} (\epsilon_{\mathbf{k}\sigma} - \tilde{\mu}) \hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k} \in \text{BZ}} [\xi_{\mathbf{k}} \hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}} \hat{n}_{-\mathbf{k}\downarrow}] \\ &= \sum_{\mathbf{k} \in \text{BZ}} \xi_{\mathbf{k}} \hat{s}_{\mathbf{k}}^z + E_a^{(\text{SC})} \end{aligned}$$

being $E_a^{(\text{SC})}$ the “anticommutation energy” coming from the kinetic term,

$$E_a^{(\text{SC})} \equiv \sum_{\mathbf{k} \in \text{BZ}} \xi_{\mathbf{k}} \quad (2.47)$$

The scalar term in the above equation is essential for estimating correctly the phase ground state free energy density. The full MFT gran-canonical hamiltonian in Nambu representation is obtained including Eq. (??),

$$\hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \hat{\Phi}_{\mathbf{k}}^\dagger h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E_a^{(\text{SC})} - \left[E_{H/U}^{(\text{SC})} + E_{C/U}^{(\text{SC})} \right] \quad (2.48)$$

(note the non-renormalized chemical potential on the left) being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta_{\mathbf{k}}^* \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix}$$

while the off-diagonal gap is a constant,

$$\Delta_{\mathbf{k}} \equiv \Delta^{(s)}$$

This hamiltonian behaves as an ensemble of spins in local magnetic pseudofields

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\operatorname{Re}\{\Delta_{\mathbf{k}}\} \\ -\operatorname{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix} \quad \tan 2\theta_{\mathbf{k}} \equiv \frac{\xi_{\mathbf{k}}}{\sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}} \quad \tan 2\zeta_{\mathbf{k}} \equiv \frac{\operatorname{Im}\{\Delta_{\mathbf{k}}\}}{\operatorname{Re}\{\Delta_{\mathbf{k}}\}} \quad (2.49)$$

precisely as was for the AF phase, with the essential difference that now the chemical potential RMP lies inside the z component of the pseudofield itself. This is a consequence of the Nambu spinors definition. We have the spin hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + E_{\text{a}}^{(\text{SC})} - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} \right] \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}} = \begin{bmatrix} \hat{s}_{\mathbf{k}}^x \\ \hat{s}_{\mathbf{k}}^y \\ \hat{s}_{\mathbf{k}}^z \end{bmatrix} \quad (2.50)$$

As was for the AF phase, to diagonalize the hamiltonian means essentially to align the spin z axis with the direction of the pseudofield. Then, we diagonalize the hamiltonian via a set of rotations similarly to Eq. (??). The main difference is that in general the pseudofield can have a y component, thus the diagonalizing matrix is made of a sequence of two rotations, both counter-clockwise, the first around the z axis by an angle $2\zeta_{\mathbf{k}}$ to place the pseudofield in the xz plane, the second around the new y axis by an angle $2\theta_{\mathbf{k}}$ to align the z axis with the pseudofield,

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}} & -\sin \theta_{\mathbf{k}} \\ \sin \theta_{\mathbf{k}} & \cos \theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \quad (2.51)$$

obtaining the diagonal matrix

$$d_{\mathbf{k}} \equiv W_{\mathbf{k}} h_{\mathbf{k}} W_{\mathbf{k}}^{\dagger} \quad \text{where} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix}$$

and the bands

$$E_{\mathbf{k}} \equiv \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$$

Finally, the Nambu spinor for Bogoliubov quasiparticles is given similarly to Eq. (??)

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} \quad (2.52)$$

Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}}^{(+)} \\ \hat{\gamma}_{\mathbf{k}}^{(-)} \end{bmatrix} \end{aligned} \quad (2.53)$$

These operators act as free fermion fields on the Bogoliubov bands $\pm E_{\mathbf{k}}$.

2.2.4 Explicit zero-temperature SC ground state at half-filling

Reasoning precisely as in Sec. ??, we can extract the half-filling zero temperature ground state, commonly referred to as the “BCS state”. Eqns. (??) and (??) hold equivalently (ignoring the σ index and extending momentum tensor product to the entire BZ),

$$|\Psi\rangle = \bigotimes_{\mathbf{k} \in \text{BZ}} [\cos \theta_{\mathbf{k}} |\downarrow_{\mathbf{k}}\rangle - \sin \theta_{\mathbf{k}} |\uparrow_{\mathbf{k}}\rangle]$$

Using now Eq. (??), we have

$$\begin{aligned} \langle \downarrow_{\mathbf{k}} | \hat{s}_{\mathbf{k}}^z + \hat{\Gamma}_{\mathbf{k}} | \downarrow_{\mathbf{k}} \rangle &= 0 \\ \langle \uparrow_{\mathbf{k}} | \hat{s}_{\mathbf{k}}^z + \hat{\Gamma}_{\mathbf{k}} | \uparrow_{\mathbf{k}} \rangle &= 2 \end{aligned}$$

so up to a phase

$$|\downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \quad |\uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}_{\mathbf{k}}^{\dagger} |\Omega_{\mathbf{k}}\rangle$$

Hence, we have

$$|\Psi\rangle = \bigotimes_{\mathbf{k} \in \text{BZ}} \left[\cos \theta_{\mathbf{k}} + e^{2i\zeta_{\mathbf{k}}} \sin \theta_{\mathbf{k}} \hat{\phi}_{\mathbf{k}}^{\dagger} \right] |\Omega_{\mathbf{k}}\rangle \quad (2.54)$$

The argument $2\zeta_{\mathbf{k}}$ of the phase factor is precisely the phase of the gap function, as can be shown easily.

2.2.5 Superconducting fluctuations suppression in the conventional HM

Let us report the SC analogous of the AF Eqns. (??) and (??):

$$\hat{s}_{\mathbf{k}}^x = \hat{\phi}_{\mathbf{k}} + \hat{\phi}_{\mathbf{k}}^{\dagger} \quad (2.55)$$

$$\hat{s}_{\mathbf{k}}^y = i \left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}_{\mathbf{k}}^{\dagger} \right) \quad (2.56)$$

while the analogous of Eq. (??) was given in Eq. (??). Let also the thermal factor

$$\text{Th}_{\mathbf{k}}^{(\pm)}[\beta] \equiv f(-E_{\mathbf{k}}; \beta, 0) \pm f(E_{\mathbf{k}}; \beta, 0) \quad (2.57)$$

and the EVs:

$$\langle \hat{s}_{\mathbf{k}}^x \rangle = \frac{\text{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \quad (2.58)$$

$$\langle \hat{s}_{\mathbf{k}}^y \rangle = \frac{\text{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \quad (2.59)$$

$$\langle \hat{s}_{\mathbf{k}}^z \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \quad (2.60)$$

Up to now, the derivation was general. However, it is immediate to show that for the pure HM the SC phase can never establish. Recall Eq. (??): explicitly

$$\begin{aligned} w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{\phi}_{\mathbf{k}}^{\dagger} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\langle \hat{s}_{\mathbf{k}}^x \rangle + i \langle \hat{s}_{\mathbf{k}}^y \rangle}{2} \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \end{aligned} \quad (2.61)$$

Now: $\Delta_{\mathbf{k}} = \Delta^{(s)} = -Uw^{(s)}$, thus if we define

$$U_c[\beta] \equiv \left[\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\text{Th}_{\mathbf{k}}^{(-)}[\beta]}{E_{\mathbf{k}}} \right]^{-1}$$

(the reason for this notation will be clear in Sec. [\[Missing\]](#)) we end with the equation:

$$w^{(s)} = -\frac{U}{U_c[\beta]} w^{(s)}$$

Since evidently $U_c[\beta] > 0$ by definition (the thermal factor is negative by construction), the above equation can be solved in $\mathbb{C}$ only by $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at this level in the HM.

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